



PINAYS STEM

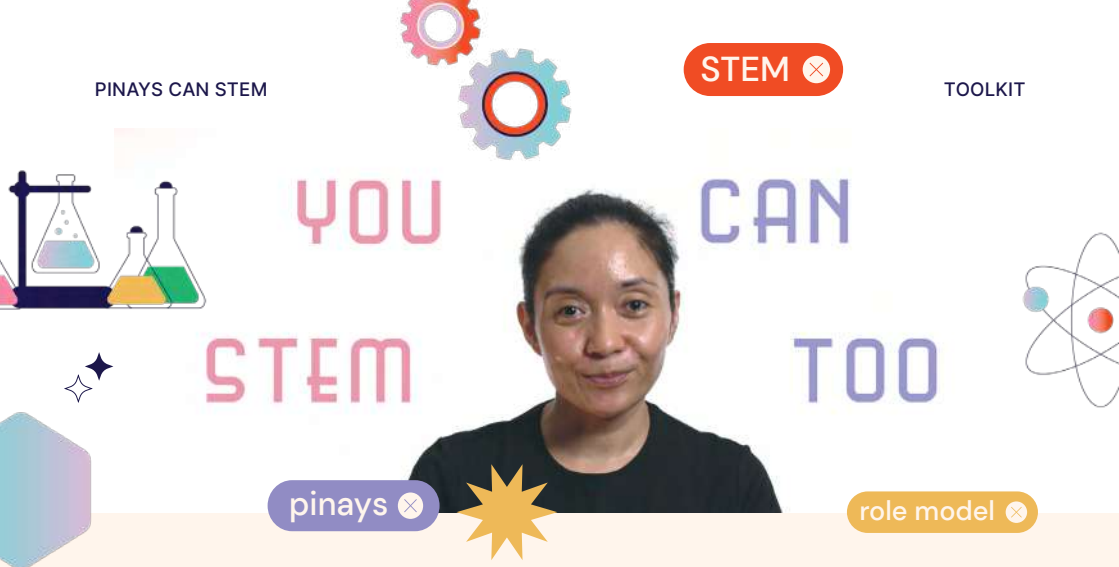
Toolkit

MADE FOR EVERY PINAY IN STEM!



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STEM in the Philippines

The Philippines ranks 16th in the world's gender gap¹, indicating the smallest gender gap in Asia. In fact, Pinays comprise 49% of the Science and Technology workforce. However, even with this growth, many factors contribute to the gender gap in Science, Technology, Engineering, and Mathematics (STEM), particularly for young female learners. On average, only 36% of STEM graduates are women², and from that, only 46.2% are employed in STEM-related fields³.

According to the Youth in STEM report (2020) entitled, *Perceptions of the Filipino Youth Towards STEM and The Need to Understand It*⁴, the presence of Filipina role models, either as STEM professionals or as industry leaders in STEM, directly influence young girls' interest in pursuing that path. There is a significant gap in the area of STEM subjects that males are more likely to pursue, such as Engineering, compared to females who are more inclined to take medical-related subjects and careers such as Chemistry and Biology. In contrast, another notable finding shows that female students across the country are more inclined to pursue STEM as a discipline if it also contributes to the society. Much is yet

to be done in raising awareness on other STEM careers, particularly on the natural sciences, technology, and mathematics.

A whole gamut of hindering factors—the male-dominated STEM education system, the lack of support towards girls' capabilities, gender stereotypes and biases, and the global digital divide—disproportionately affects women and girls everywhere. In the country, the apparent lack of female role models, prevailing gender stereotypes, and the underrepresentation of women professionals in STEM discourage young Pinays to pursue the STEM track.

"I am a consistent honor student and pangarap ko talaga na magkaroon ng M.D.-Ph.D sa dulo ng pangalan ko. A future doctor that'll soon make a great change. But a lot of people around me are insisting that I should not continue because it's a waste of time and money. Sayang daw ako and so on. Despite these comments, I still chose STEM as my strand with no hesitation and I'm proud of being a STEM student."

- A testimonial from a Pinay in one of Pinays Can STEM's posts.

gender

Celebrating every Pinay's skills, capabilities, and dreams in STEM

STEM+PH is the flagship program of Unilab Foundation, which promotes STEM education by nurturing integrated STEM learners, innovators, and the workforce. To help realize this, STEM+PH launched Pinays Can STEM (PCS), an advocacy platform and an online community that aims to empower and encourage young Filipinas in STEM. Pinays Can STEM encourages and influences Filipinas to pursue their dreams by providing them with an avenue to connect with like-minded Pinays that will inspire them, and to explore opportunities that gives them access to right tools and experiences that will empower them in their decision-making. Truly, a community of STEM champions can help propel the Pinays in STEM in leading the future and achieving greater heights.

Breaking barriers and defying gender perceptions, Pinays Can STEM provides insightful infographics, interactive quizzes, local and international inventions and achievements, and inspirational role models for all young Pinays. Through informative social media cards, articles, and videos centered around role models and empowering young girls and women in STEM, Pinays Can STEM continues to engage more Filipinas to pursue the STEM track. Relatable and well-curated contents include the discussion of inadequate support from their environment, gender discrimination, lack opportunities in STEM—all of which affect their own choices in their careers. In just a year, Pinays Can STEM has featured more than 30 Pinays with an audience reach of over 700,000, and more than 4,000 Facebook and Instagram pages followers.

"In the future, I envision women to be and feel as welcome in STEM as in any other discipline."

- Angelina Aquino, Lecturer under the Electrical and Electronics Engineering Institute at University of the Philippines, in Pinays Can STEM's Pinay of the Month article 'Defying Deviations: Exceeding the Norm with Angelina Aquino' last December 2020.

PiNAYS
can STEM



opportunities

Achieving gender equality in STEM

IN PARTNERSHIP WITH

Philippine
Commission
on Women



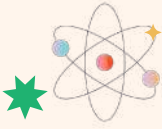
Pinays Can STEM recognizes to build an inclusive community for women to grow in STEM, their surrounding environment should actively take part in the process. Institutions such as schools and workplaces can promote gender equality by creating inclusive leadership opportunities for women leaders, and institutionalizing a culture of open and safe spaces that cater to varying needs of women. Male classmates, peers, and family members, can be rallied upon as allies who commit to promoting gender inclusion and sensitivity. Leaders can also strengthen and implement existing policies that facilitate STEM education and fields, promote women empowerment, and strengthen partnerships in the women's sector, policies against harassment, and accountability mechanisms.

To further strengthen the gender advocacy in STEM in the country, Pinays Can STEM has partnered with the Philippine Commission on Women (PCW), a government agency run by the government of the Philippines that aims to promote and protect the rights of Filipinas. Centered around the belief that women deserve equal opportunities, PCS and PCW are working towards having more

women who are qualified, confident, and engaged in STEM-related careers not only in healthcare, but also in research, engineering, information technology, and public health. These objectives respond to the findings of the recent study by the International Labour Organization (ILO) underpinning that gender diversity in the workplace has increased profitability, productivity, and creativity in various Filipino companies, indicating the growing need to invest in building opportunities for women in STEM⁵.

By exposing our girls to STEM, we give them the freedom to explore its wonders, opportunities, and eventually develop their pathways to STEM learning and STEM careers. PCS and PCW recognize the youth's role in transforming society and, as the leaders of the future, it takes conscious effort to plant the seeds on the importance of realizing the women's full potential to actively participate in driving the economy, and of institutionalizing gender equality and inclusive leadership opportunities in the workplace. Together, we hope to influence young Filipinas to start with the small things, then further transform their STEM dreams into reality.

Welcome to your



STEM journey!

This digital toolkit provides content and resources, ranging from stories, tips, and other informational materials, to inspire young Pinays interested in STEM such as you! Through the toolkit, Pinays Can STEM gives you the freedom to **explore the wonders and opportunities in STEM** and eventually **develop your passions in your learning and careers**.

What's in the toolkit?

Click on the boxes to check out each section!

Kickstarting your STEM Journey

For the curious Pinay, this is a guide on the different STEM careers that you can explore! This contains information and trivias on the STEM track, content on STEM careers and women empowerment.

Pinays Can Do Anything!

Learn more about Pinays who are changing the world, and at the same time, making their dreams come true. From a collection of STEM journeys, to Filipina achievements and inventions, these role models are here to tell you that you can STEM too.

PCS TV

Tune in and watch our awesome Pinay role models in STEM present the wonders of their work and their journey to achieve it. You will also see an interaction and exchange of ideas between students and their mentors.

The STEM Library

Are you interested to know how STEM is important in everyday life? Check out our collection of reading and learning resources from our library.

STEM Directory

Get to know other STEM and women organizations in the Philippines and abroad!

Pinays Can STEM Slambook

Pinays Can STEM is here for you! This slambook is where you can start your STEM journey by listing your goals and aspirations in STEM, as well as other learnings and inspirations from the contents of this toolkit.

- ✦ The Slambook is an interactive file. You may open it using any PDF application, software, or through Google Chrome, as long as you are able to fill up or edit the file.
- ✦ If you'll be opening through Chrome, make sure to save your changes before closing the PDF (Check the download arrow beside the printer icon, upper right corner, click save or "With Your Changes").





Kickstarting your
STEM Journey

What is STEM?

What is the first thing that enters your mind when you hear STEM? How do you envision your STEM story?

STEM, or Science, Technology, Engineering, and Mathematics, is a broad term used to group these academic disciplines. Subjects such as Earth Sciences, Calculus, Physics, Algebra, are also under this strand! STEM now plays a vital role in the Philippines' economic development, especially as we move closer to the 4th Industrial Revolution. Rather than teaching them separately, STEM integrates them into a cohesive learning paradigm based on real-world applications.

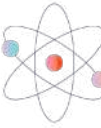
According to our Youth in STEM report, only 52% of Filipinos are currently taking up STEM in university, yet 80% recognize the positive impact of STEM to society and acknowledge the importance of STEM skills in acquiring a good career in the future. Given that most STEM jobs are now in high demand, students still feel that they lack the resources to capacitate and equip themselves with the necessary skills for the future of the workforce.

No field is for 'boys only' (nor for 'girls only'). If it's your passion and you can see yourself growing and developing your skill in STEM, go for it!



Sabrina Ongkiko
Science Teacher at Culiati Elementary School

There is now a need for more people in the field to innovate and build a better Philippines with STEM. STEM can create transformative opportunities not just for corporations, but also for individuals by equipping them with in-demand skills that could lead to better opportunities and a means to deliver impact. For many, the hunger for a STEM career is fueled by a deeper passion, dream, or call to purpose. And this could be you!



KICKSTART YOUR JOURNEY

STEM in Senior High School

Exploring the world of Science

Decoding the language of Tech

Gearing up for Engineering

Finding the x, y, and love for Math

Bonus content!

Best in Class

Preparing SHS students for STEM

Let's face it—school is hard enough as it is right now. As the COVID-19 pandemic adds pressure to students, it can be extra challenging to pursue one's dream course and path. We paired up with a few study buddies from the non-profit organization called Kababaihan Para Sa Siyénsiya (@siyensiya.ph on Instagram) to take us through their personal STEM syllabus and their testimonies for the Students today, Scientists tomorrow.

[SHARE THE BLOG](#)

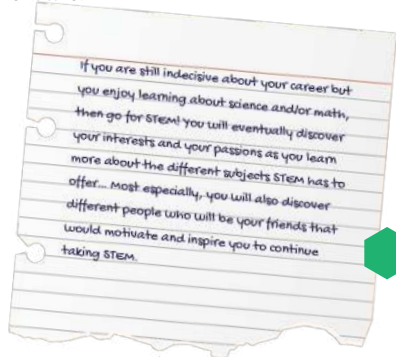
'In STEM, learning does not stop'

As a Medical Laboratory student, Kyla's first memory of STEM was back in 6th Grade when they learned about the different body systems. Skipping ahead to the future lessons, she soon filled her textbook with her own notes and highlights. A turning point in Kyla's journey was actually seeing a specimen slide during one of her Biology classes, where she realized that there's more to life (and STEM) than visible to the naked eye.

You don't have to know everything to be in STEM. STEM is indeed a challenging track, and it is absolutely normal to not know all the answers to the questions. In STEM, learning does not stop. You continually learn, discover, and develop your knowledge. So keep wondering, and keep asking questions. In fact, these questions may turn out to be something that you will be able to answer in the future - and in the process, be able to contribute to the greater good of society.

'Nothing is challenging when you're passionate'

Denabea started her love for STEM through Mathematics. The decisiveness of computations and numbers has always been 'satisfying' to the young Thomasian. She is now beginning to take a keen interest in biology and plans to be a doctor in the future.



'Do not let that fear take over you'

On the cusp of her STEM journey, Feaid has taken a multitude of electives (from Agriculture to Computer Science) to prepare herself for her dream course of Agricultural Chemistry. She understands that everything that is happening in the world, such as the African Swine Fever, Taal Volcano eruption, and the COVID-19 pandemic requires more people who will step up for the challenge of delivering frontline medical/health services.



Let's admit it: STEM is hard. It is complicated. I get why most people are very intimidated by it. But do not let that fear take over you. STEM has a lot more wonderful things other than hard concepts and computations. Let those wonderful things drive you to enjoy what you learn. The cycle of questioning and answering these questions would help you to learn more and more. There are a lot of inspiring and fun people in this field. They are not the stereotypic nerds that are serious all the time. Rather, they are passionate about what they do and they lovingly share that knowledge in a very, very fun way.



'Always make sure you create for good'

Jammy is lucky enough to be surrounded by family members who are in the STEM field. With an inkling for Mathematics, she decided to join after-school classes and various competitions to prepare herself for high school STEM subjects. After landing an internship with a local pharmaceutical company, Jammy now wants to take up chemical engineering to improve the country's healthcare industry.

STEM can seem daunting at first, but it is definitely worth it! Whether it be discovering a new vaccine or bringing clean water to local areas, STEM plays a vital role in solving problems at a local or global scale. Regardless of the impact you choose to make, always make sure that you create and innovate for good. The goal of giving back to the people and improving the lives of your community is what makes STEM worth doing.



'You don't have to always be 100% sure'

As early as Grade 7, Bree knew she wanted to take up STEM. Aside from focusing on school, the idea that STEM can be used to better the lives of so many people keeps Bree going. Her ultimate goal is to make STEM "for the people".



If you are still indecisive about your career but you enjoy learning about science and/or math, then go for STEM! You will eventually discover your interests and your passions as you learn more about the different subjects STEM has to offer... Most especially, you will also discover different people who will be your friends that would motivate and inspire you to continue taking STEM.





'Open your eyes to reality'

Before taking up BS Biology major in Medical Biology, the STEM 'adventures' sparked Katrina's interest in the field. From 8th grade science investigatory projects (SIPs) to representing her region in Marikina and Baguio during DepEd science fairs, stepping outside of her comfort zone continues to push Katrina to her goals.

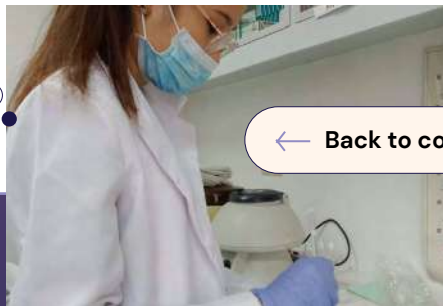
Get out of the four corners of your classroom (or in this case, your laptop screen) and open your eyes to reality. Science-related courses, despite being seen as an intensive course in college, are still undermined in our society today. With the way things are handled this pandemic, it is no doubt that scientists are much more needed now to create data-based and scientifically backed-up solutions for the country. It is completely understandable to still be hesitant about it, which is why you should join activities and organizations outside of your comfort zone and maybe it might work for you!



'There will be a sense of fulfilment'

As a current Physical Therapy student, Laysa has always wanted to be a doctor. Even though the end goal has always been clear, Laysa has discovered lessons about the world that's gone beyond her expectations. If she could describe her STEM journey in one word, it would be 'electrifying'.

Always open your heart to what really makes you happy as it can be really beneficial for you in the long run. At the end of the day, there will be a sense of fulfillment blooming in your heart as you remember that you are there not just to make your loved ones proud, but to serve your country as well. This country still has a huge room for improvement in terms of STEM-related aspects thus I hope this makes you determined to become part of the change. Padayon, future Pinay sa STEM!



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Exploring the world of Science

A career in science, just like every other career in STEM, is designed with an end goal of improving people's lives and ultimately, creating a better world to live in. On this pursuit, STEM learners develop their knowledge and understanding of the world around us. Whether it be in medicine, research, or the environment, we're here to help you explore your opportunities to help you grow as a student and professional!

ARE YOU READY TO START YOUR JOURNEY IN SCIENCE?

Find science in everyday activities

Science is everywhere and what makes it so great is that it keeps you curious. Explore your surroundings, don't be afraid to ask questions, and see science as a never-ending quest for answers.

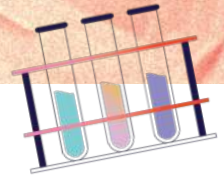
Research on opportunities

Science is a very diverse field! From medicine and research, education and food science, and many more explorable fields of study, find your place and cultivate your interests.

Remember: You don't always have to be the best

Science can be hard for some. Getting the lowest grades, or even failing, doesn't mean your science journey stops there. See every encounter with science as a chance to grow, and continue to discover its applications in real life to help you better understand it.

CAREERS



Chemist

Does chemistry excite you? Are you interested in the chemical composition, structure, and properties of substances? You're probably a chemist in the making! A chemist is a scientist who researches chemical substances, performs experiments with the properties of chemical substances, measures the effects of chemical compounds in various situations, and studies inter-chemical reactions.

Astronomer

Astronomers explore and understand the universe and what goes on within and beyond it. Through satellites, telescopes, and other space equipment, they are able to collect data in order to develop theories about the stars, planets, and galaxies.

Environmental Scientist

With the effects of climate change in our environment, it's no surprise that more people are showing interest and becoming advocates. Environmental scientists identify health problems, causes, and possible solutions. They help increase our understanding of the world through environmental research, experiments, and by developing policies and practices.

Epidemiologist

When disease outbreaks or other threats emerge (such as the COVID-19 pandemic) you can definitely count on epidemiologists! They determine how to control, stop the spread, or prevent it from happening again. Epidemiologists investigate causes and patterns of disease and injury in humans, and reduce these risks through research, education, and health policy.

Zoologist

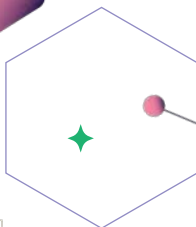
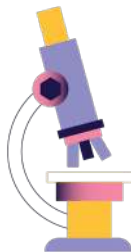
Humans aren't the only living beings in this world! Zoologists study the origins, genetics, and behaviors of animals, either in their natural habitats or in captivity (zoos or aquariums), which is key in the preservation of species and protection of wildlife.

Botanist

Plants are vital to everything that exists on earth. Botanists study various aspects of plants such as the evolution, reproduction, processes, and relationships with the environment and other forms of life.

Agricultural Technician

Agriculture plays a significant role in the Philippine economy. Agricultural technicians work in the forefront by helping scientists conduct experiments. As the world's population grows larger, so does the need to improve the quality and yield of food crops and animals food sources.



Pinays in Science



Mary Suzette Angeles

Mary Suzette Angeles is a research specialist at the Molecular Diagnostics and Cellular Therapeutics at the Lung Center of the Philippines. She led the team of scientists responsible for developing a Circulating Tumor Cell Analysis (liquid biopsy), a non-invasive sensitive test that detects residual cancer cells in the body. This helps clinicians identify better suited treatments for cancer patients and lays down the research for targeted chemotherapy to lessen the negative side effects of cancer treatment.

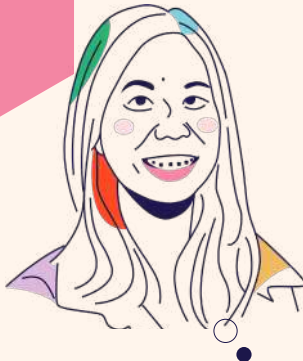
Cheryl Soledad

Cheryl is currently a Lead Food Designer at The Kellogg Company's Natural & Insurgent Category – Research & Development Team, leading product development projects for the Kashi brand. Prior to joining Kellogg's, she was an Associate Principal Scientist at General Mills, Inc., where she spent most of her time in research and development across several product categories. She earned her MS and PhD in Food Science from The Ohio State University, and her bachelor's degree in Food Technology from UP Los Baños.



Erina Molina

Erina Molina is an environmental scientist, conservationist, and a National Geographic Young Explorer. She engages with local communities in the conservation of marine resources to bridge the gap between science and the community. She has worked as a conservation research specialist at Haribon Foundation, where she shared her knowledge to help Filipino communities identify vulnerable or locally extinct reef fish species. In 2015, she also co-produced a virtual reality film entitled Grandpa's Reef, which provides an immersive experience to its viewers about the underwater world.



Andreia Carrillo

Andreia Carrillo is an astrophysicist, known for leading the team that discovered a new dwarf satellite, dw1335-29, of the Messier 83 group (M83, also known as the Southern Pinwheel Galaxy) located some 85,000 light years from its host. She aims to foster a deeper appreciation on science and astronomy for the youth by providing opportunities in the Philippines. Other than her work in STEM, she is also an advocate for education where she has participated in numerous teaching and outreach activities.

**Dr. Merlinda Palencia**

Dr. Palencia is a chemical engineer known for developing 'Organo Minerals', a powder mixture made from naturally occurring minerals. Studies showed that the mixture was able to improve the water quality of Adamson University's septic tank. Seeing the potential of her invention, she went on to apply the technology for intellectual property rights and later marketed it as "Vigormin".

**Oona Nery**

Oona is a registered chemist and Science Research Specialist II at the Philippine Nuclear Research Institute (PNRI) – International Cooperation Section. Prior to this, she was a Chemist at PNRI's Agriculture Research Section where her work focused on the use of nuclear and isotopic techniques in nutrient management, soil erosion assessment, and plant mutation breeding.

**Marielle Eduardo**

Marielle is an astronomer at the Institute of Astronomy and Astrophysics, Academia Sinica in Taipei, Taiwan. As part of the Planetary Science group, she searches for and characterizes objects beyond the orbit of Neptune, or the "Trans-Neptunian Objects (TNOs)", using images obtained from the Subaru Telescope in Mauna Kea, Hawaii. She is also a Lecturer in Physics at UP Baguio where she teaches fundamental physics laboratory courses, computational physics, and solar system astrophysics to undergraduate students.



Decoding the language of Technology

Pursuing a career in tech is an exciting industry that continues to grow, and often consists of people from a variety of backgrounds working together. Given that the field is full of opportunities, there is no doubt you will be able to learn new skills or gain more experience in different areas. Learn more about these awesome careers in tech!

ARE YOU READY TO START YOUR JOURNEY IN TECH?

Research, research, research!

Tech is a very broad field and you can come from any background and land a career here! Familiarize yourself with the different kinds of jobs in tech, look for tools and resources online, and find out where you can see yourself grow. After doing the necessary research, take a moment and decide on a tech career path that you want to pursue.

Don't be afraid to get started

It's perfectly fine to start from scratch and there will be times you might even make some mistakes along the way. Whether you're learning how to code or designing a website, starting your career in tech will lead you to unexpected paths, but what matters the most is to prepare yourself for the changes along the way! Deciding to kickstart a career in tech is great, but putting in the necessary work to learn is even greater.

Take part or build your projects

Now that you've built the skills and a good foundation of knowledge in tech, start taking on projects. Having this experience lets you improve your tech skills in ways you can't imagine! You can also join organizations and

CAREERS



Data Scientist

Data science has become a popular career nowadays, especially given the demand to make sense of all data and information in the world. Data scientists spend most of their time writing algorithms and analyzing data while developing innovative approaches to predict how something will behave based on the recorded data.

Computer Programmer

A computer programmer mainly writes the code that helps software applications operate more efficiently. People in the field design and update software solutions, write and update source-code, and manage different operating systems.

Web Developer

To simply put it, web developers design and build websites! They take charge of the appearance of a website and its technical aspects, such as the site's loadingspeed, visitor traffic, and many others. Some web developers also take charge of copywriting and the creation of the content that requires technical features.

User Experience Designer

User experience design is also one of the most in-demand jobs especially in the creative and tech industry! To differentiate from web developers, UX designers are responsible for how a product or website feels. It is their responsibility to focus on and put the users' emotional and functional needs at the center, and use these insights to create or improve ease of use of any app or website.

IT Technician

If you enjoy working with technology and are interested in knowing more about the industry, a career as an IT technician is a great start! IT technicians install, troubleshoot, and fix the hardware and software in a computer system. They can also go on to more specialized roles like systems engineer, network administrator, or cloud administrator.

Artificial Intelligence Professional

Are you familiar with Siri, chatbots, self-driving vehicles, and facial recognition softwares? If you find these cool, you might want to start a career in Artificial Intelligence (AI)!

Artificial intelligence is all around us, and it continues to be a growing form of technology in the modern world. As defined, AI refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions. What's great about the field is that it spans across different industries such as healthcare, technology, media, financial services, marketing, government, gaming, agriculture, and many more!



Pinays in Tech



Chiara Ledesma

Chiara is a Data Scientist (Machine Learning Team) at Thinking Machines. She graduated from Ateneo de Manila University with a degree in Computer Science, where she was one of the 73 chosen students for the Google Women in Tech scholarship and beat out 25,000 applicants across South East Asia.

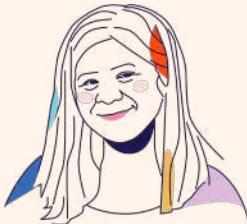
Isabel Sieh

Isabel is the President and Founder of Girls Will Code, an organization that encourages girls to participate in activities related to programming, robotics, and engineering. Isabel founded Girls Will Code when she was only 10 years old!



Chia Amisola

Chia is the founder of Developh, an international student-led organization that nurtures innovation in the youth through accessible technology opportunities. Chia founded Developh in her junior year of high school. She is now a student at Yale University.



Cara Wilson

Cara is the co-founder and Executive Director for the For the Women Foundation (FTW), a non-profit organization that aims to change women's lives by offering free data science and AI training for future leaders like herself. More than just improving the technological literacy and skillset of women, FTW Foundation has taught their Filipina scholars to believe in themselves and strive for a better future.



**Mylene Abiva**

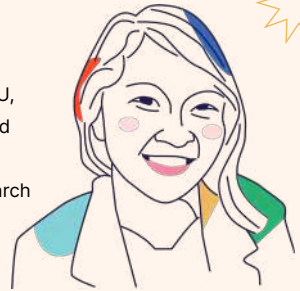
Mylene is the President and CEO of FELTA Multi-Media Inc., where she oversees the operations of the company. Since 2000, she has been paving the way for students to gain different opportunities in engineering and Robotics in the Philippines and abroad. She spearheaded the Philippine Robotics Olympiad in 1999 to further promote Robotics and reach schools all throughout the Philippines. In 2001, in their very first competition at the World Robot Olympiad, the Philippine national team won a silver medal. And in the 2012 at the same competition, the Philippines sent an all-girls team become the first all-girls team to win the gold medal! This accomplishment was a first for the Philippines and the world!

**Amber Teng**

Amber is a Data Scientist and the Co-Founder of KadaKareer, a community-based career development platform for Filipino students launching their digital careers. Amber stumbled upon data science by accident. Back then, she didn't even know what data science was nor did she have any prior experience into coding and tech. After obtaining her bachelor's degree in Economics and Archaeology at Brown University, she then pursued and earned her MS in Data Science from NYU Center for Data Science.

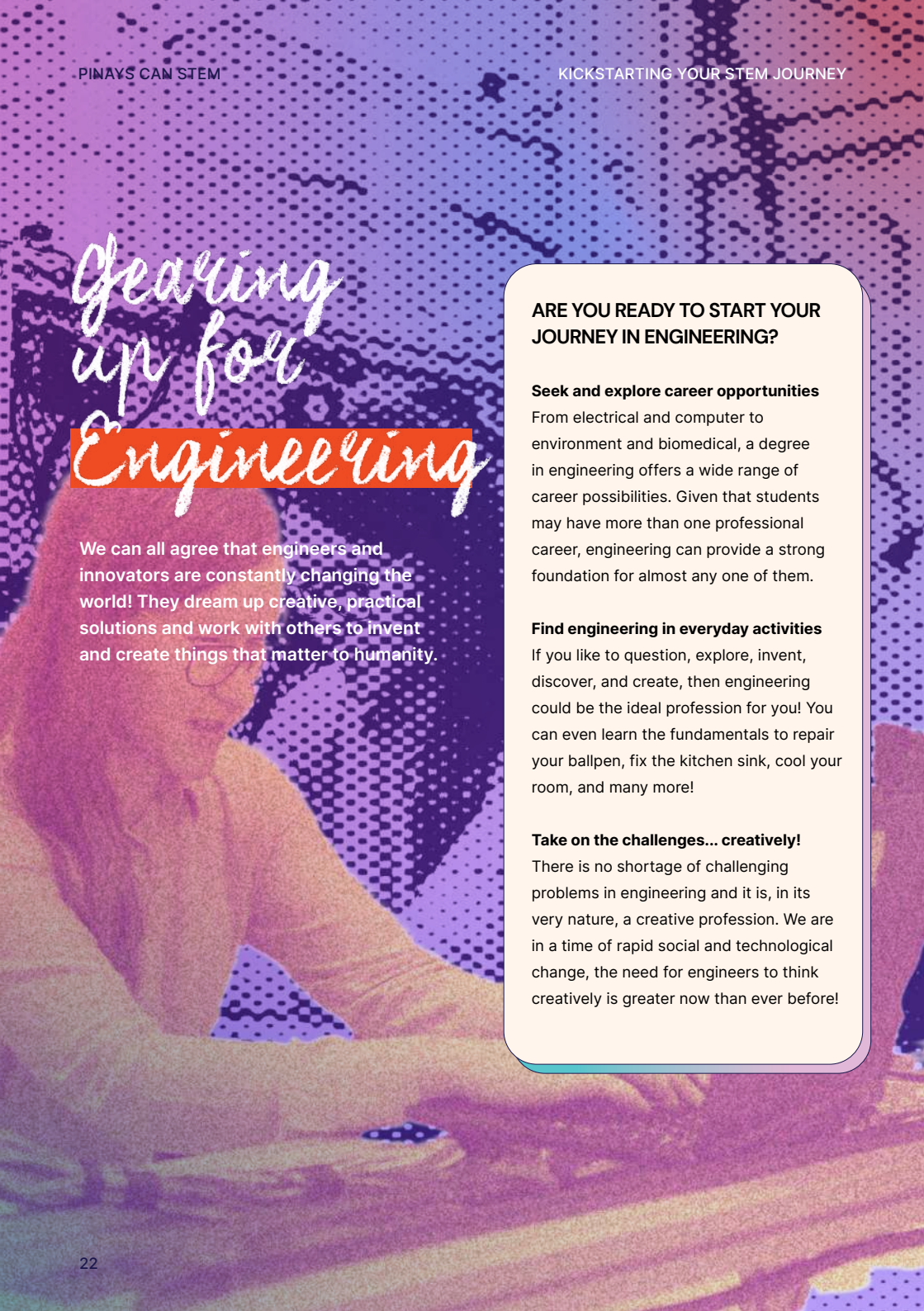
**Lian Castillo**

Lian is an IT professional with a bachelor's in Computer Science at DLSU, and a masters in Information Studies at the University of Tokyo. She had previously handled professional projects in culture-based conservation and blended learning. She is currently based in Chiba, Japan as a research student in Sustainability Science at UTokyo's Kashiwa campus.

**Cristina del Rosario**

Cristina is a Filipina UX designer. Back in college, she created a vaccination mechanism which made it easier to administer vaccines in rural areas of the Philippines. This led her to become a finalist for the International Design Excellence Awards, and the syringe was later patented by a biomedical company. She also helped develop an application, SariLoad, that aimed to improve the technological and financial literacy of sari-sari store owners.





Gearing up for Engineering

We can all agree that engineers and innovators are constantly changing the world! They dream up creative, practical solutions and work with others to invent and create things that matter to humanity.

ARE YOU READY TO START YOUR JOURNEY IN ENGINEERING?

Seek and explore career opportunities

From electrical and computer to environment and biomedical, a degree in engineering offers a wide range of career possibilities. Given that students may have more than one professional career, engineering can provide a strong foundation for almost any one of them.

Find engineering in everyday activities

If you like to question, explore, invent, discover, and create, then engineering could be the ideal profession for you! You can even learn the fundamentals to repair your ballpen, fix the kitchen sink, cool your room, and many more!

Take on the challenges... creatively!

There is no shortage of challenging problems in engineering and it is, in its very nature, a creative profession. We are in a time of rapid social and technological change, the need for engineers to think creatively is greater now than ever before!

CAREERS

Sound Engineer

Mixing technical knowledge with creativity, sound engineers can produce high-quality sounds for different contexts such as live events, movies, TV shows, podcasts, radio, advertising, gaming, and many other media purposes.

Gameplay Software Engineer

Video games and animation have become more popular over the years. There is now a demand for more engineers and programmers who specialize in physics and artificial intelligence to create realistic and unique animations.

Industrial Engineer

Industrial engineers are in-charge of evaluating and improving production processes, making systems more efficient. From evaluating production schedules and process flows, they determine ways to improve efficiency and develop management control systems for improved effectiveness and lowered costs.

Biomedical Engineer

Biomedical engineers work closely with biological and medical scientists to design and develop equipment, systems and software for the healthcare industry. They are mainly responsible for installing, maintaining, evaluating and repairing biomedical equipment.

Civil Engineer

Civil engineers create, improve and protect the environment in which we live. They plan, design and oversee construction and maintenance of

building structures and infrastructure, such as roads, railways, airports, bridges, water and sewerage systems.

Mechanical Engineer

Mechanical engineers use STEM concepts to design, develop, and test mechanical devices. Their responsibilities include designing effective tools, improving mechanical technologies, drafting blueprints, constructing prototypes, testing theories and proving functional ways to use products and procedures.

Electrical Engineer

Electrical engineers mainly work with power generation and supply systems. They regularly evaluate existing systems and solutions and look for ways to improve efficiency in the energy sector.

Aeronautical Engineer

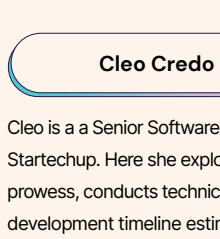
Aeronautical (or also aerospace) engineers typically work in aerospace, aviation, and defense. They develop, design, test, and maintain aeronautical engineering systems that are used in helicopters, space shuttles, passenger airliners, fighter planes, stealth jets, and other airborne vehicles.

Pinays in Engineering



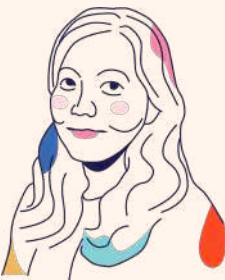
Maria Kathrine Co

Maria is currently the Supply Chain Commercial Lead in Shell Business Operations - Manila, where she supports the Manufacturing site in Singapore. She is responsible for overseeing the operational and tactical contracts for the Engineering, Maintenance, Services, and Disposal Categories.



Cleo Credo

Cleo is a Senior Software Engineer, and currently the Chief Technology Officer of Startechup. Here she explores different technologies to leverage the company's technical prowess, conducts technical assessments on software systems, comes up with project development timeline estimates, and performs code reviews with the engineering team.



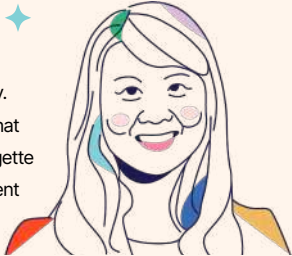
Claire Pascua

Claire is currently a PhD candidate at the University of Auckland (New Zealand), where she specializes in structural and earthquake engineering. Her PhD thesis is focused on the seismic performance of buildings with combined concrete walls and steel frames. Her research involves numerical modelling of such buildings and experimental tests on connection details to understand how they will perform during earthquakes.



Anne Brigitte Lim

Brigitte is a solar energy engineer and completed her masters from Arizona State University. She won the UN's 2017 Geneva Challenge competition for her team's Solar N3E invention that aimed to boost employment in the country by training workers in the solar energy field. Brigitte is currently working with the UN on a project called "DREAMS" which stands for Development for Renewable Energy Application Mainstreaming and Market Sustainability.



Anne Aaron

Anne Aaron is a Filipina engineer and the director of video algorithms at Netflix! She is responsible for the software that enables more than 86 million Netflix members worldwide to watch streaming videos with the best possible quality. She is recognized as one of the 43 most powerful female engineers of 2017 by news website Business Insider and was featured among "America's Top 50 Women In Tech" in 2018 by Forbes Magazine.

**Angel Palma**

Angel Palma is a Mechanical Engineering student and the inventor and founder of AirDisc Cooling Technologies. When she was only Grade 10, she "accidentally" came up with her revolutionary AC invention that uses a disc-shaped compressor. Her air conditioner does not use chemical refrigerants that harm the environment. Instead of using a cooling medium, Palma's AC uses low compression and a high volume of air molecules as refrigerant replacement. She is recognized by the International Federation of Inventors' Association (IFIA) for inventing her "AirDisc" and winner of the 2019 James Dyson Award.

**Tetchi Cruz-Capellan**

Tetchi Cruz-Capellan is the CEO of Philippine renewable energy provider SunAsia Energy Inc. and founder of the Philippine Solar Power Alliance (PSPA). She is the country director of a rural electrification project, Alliance for Mindanao Off-Grid Renewable Energy (AMORE), funded by the United States Agency for International Development (USAID). Her leadership was responsible for providing energy to the remote and poverty-stricken villages in the island of Mindanao, Philippines. As a result, 6,000 homes and 500 community facilities received electricity through solar and hydro-power.

**Maricor "Jing" Soriano**

Jing Soriano is known to develop Teardrop: a low-cost, coral reef imaging device consisting of a Banca-towable hull that can hold an underwater video camera, software that stitches images, and a portable GPS locator. The Teardrop can create videos showing shallow-water coral reefs. It has been donated on over 13 locations nationwide and is currently being adopted by the Department of Environment and Natural Resources for its coral reef management program.



Finding the x, y, and love for Math

The logical thinking, problem-solving and decision-making skills you learn while studying a mathematics degree are highly valued by employers across many job sectors. Math is also all around us and it helps us understand the world better. More than just numbers, math is actually important in our everyday lives! From baking and cooking, sports and strengthening, shopping and spending, and all the way to arts and crafts, you can surely find the math in everything!

ARE YOU READY TO START YOUR JOURNEY IN MATH?

Learn from real-life applications and scenarios

It's time to go beyond your textbooks. Math is beyond finding the 'x' in your tests or calculating the number of apples you bought in a day. The subject is very useful in our day-to-day lives and improves our ability to analyze, solve problems, and think critically.

Use your math skills in games!

A common misconception is that math is all about the numbers. In reality, math actually involves logic, organization, and analysis, which are also found when we play games that involve logical and arithmetic skills. What's great about games is that it helps reduce that math anxiety we all feel and experience!

Appreciate your mistakes

It is no doubt that we have made make a lot of mistakes in our classes. This often makes us discouraged to even continue. However, these setbacks can be helpful in the long run, as they will help you learn and do better. This is an important skill we all need.

CAREERS

Financial Manager

Financial managers create financial reports, direct investments, and plans and strategies for the long-term benefit of a business or an organization. Being in the field also means you get to work in a variety of areas, may it be in hotels, banks, insurance, and many more!

Operational Researcher

Using the skills obtained in math, Operational researchers use analytical methods to assist people in developing better systems and operational procedures by examining operations and using mathematical modeling, computer software, or other analytical approaches to find more effective ways of working.

Meteorologist

Using information obtained from the land, sea and upper atmosphere, Meteorologists predict the weather and study the causes of weather conditions. They use computerized, mathematical models to forecasts concerning weather and climate patterns.

Investment Analyst

Investment analysts help traders, fund managers and stockbrokers make the best decisions with regards to investments. With the experience they provide, they make sure that investment portfolios are well-managed, and that potential investment opportunities are highlighted.

Statistician

Statisticians have continued to play a significant role in research and academia, and over the years, even in the businesses. Statisticians apply statistical methods and models to real-world problems by gathering, analyzing, and interpreting data to aid in business decision-making processes. Statisticians work in many different industries, and such as business, health and medicine, government, physical sciences, and environmental sciences.

Risk Analyst

Risk Analysts use advanced analytical skills to calculate the risk of a financial decision by reading and analyzing financial data, creating visual models to represent possible outcomes and preparing reports about business decisions.



Pinays in Mathematics



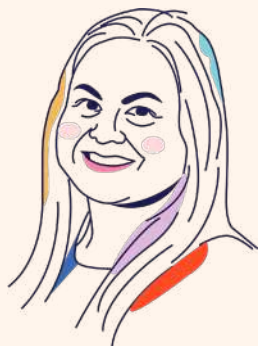
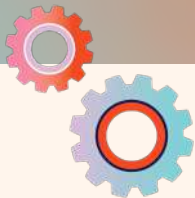
Dr. Kathleen B. Aviso

Dr. Aviso is an Associate Professor in the Chemical Engineering Department of De La Salle University. Through mathematical models, she strives to make the world a better place by tackling national and global concerns such as climate change, efficient water use, and carbon dioxide capture and storage.

Dr. Marian Roque

Dr. Roque is a mathematician and is recognized for her works in the field of mathematics, specifically the theory of ordinary and partial differential equations. Thanks to her creative ways of mentoring, she has inspired many of her students to appreciate mathematics and become scholars, who eventually joined the academe or are mathematical practitioners themselves. Because of her brilliance, she has also elevated the status of Mathematics in the Philippines and greatly influenced the professional growth and development of Filipino math teachers and students.

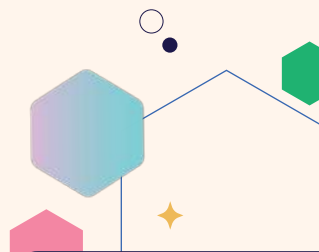


**Michelle Soledad**

Michelle is currently an electrical engineer and educator. She focuses on research and teaching fundamental engineering courses, with roles at Virginia Tech and The Ohio State University. She is also the Assistant Professor in the Iron Range Engineering (the Bell Program of Minnesota State University, Mankato) where she teaches upper division electrical engineering courses. She used to be a software engineer for Accenture, and was also the Assistant Professor and Department Chair for Electrical Engineering at the Ateneo de Davao University before pursuing doctoral studies.

**Dr. Ninette De Las Peñas**

Dr. Ninette is the Associate Dean for Research and Creative Work of the Loyola Schools, Ateneo de Manila University. Her most recent project was leading a research team to develop a mobile app for math education called “Mathplus”. The project aims to instill critical thinking skills in children grades one to eleven, while also aiding in the transition to online and blended learning amid the current health crisis. She has also used the language of mathematics to communicate other topics of interest, as she is best known for her work on woven mat patterns. Alongside two co-authors, she talks about the symmetry of mats from Philippine indigenous groups using the language of mathematics in their work called “Weaving Mat(h)s”.

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**BONUS
CONTENT!**

Fascinating and Wonderful
**STEM jobs you
probably didn't
know about**

SHARE THE BLOG

In case you didn't know: STEM is pretty much everywhere! Anywhere we turn, some facet of Science, Technology, Engineering, or Math have made our lives the way it is now. But in fact, there's more to the field than the usual jobs that come to mind.

In case being a scientist or a doctor isn't for you, we're here to list down some of not-so-common STEM careers that may just be your unsung calling.



Cosmetics Scientist/Chemist

Ever wonder who gave the green light to your fave makeup? Look no further than a Cosmetic Scientist or Chemist! These specific types of scientists develop and perform trials on makeup, toiletries, perfumes, and beauty products we use on a regular basis.

Though they may not be part of our vocabulary now, the number of cosmetic chemists is expected to grow in the workforce between 2016 and 2026.

How can I start?

Cosmetic Scientists/Chemists are likely working on R&D teams of cosmetic companies. If you're looking for a specific course, Centro Escolar University in Manila is the first and the only university that offers BS Cosmetic Science in the Philippines.



Food Technology

With food security in the country turning into a food insecurity, the art of learning how to grow food in the modern world is a must. Food technologists combine modern tech and food science to the process of selection, preservation, packaging, and distribution of safe food for everyone!

With the Philippines being abundant with natural resources, food technology can be a key process in helping our farmers bring sustainable food to the table.

How can I start?

A Food technologists' related fields include analytical chemistry, biotechnology, engineering, nutrition, quality control, and food safety management. The University of the Philippines also has a 4-year program in BS Food Technology.



Forestry and Agriculture

Just like Food Tech, you don't have to look far to find STEM's benefits in agriculture and our natural resources.

Do you remember the typhoons Rolly and Ulysses? It's widely debated that a thriving Sierra Madre mountain range would have prevented the extreme floodings. That's where the role of STEM comes in, as the country's forestry and agriculture need scientific data to shift public attention to what's really happening.

How can I start?

The Department of Environmental Resources has its own Forest Management Bureau (FMB), with positions like Forest Management Specialists, Information Analysts, and more. There are also numerous universities offering programs with a BS in Forestry.



Clothing Technology

Did you know that fast fashion is hurting our environment everyday? In exchange for fast and cheap clothes churned out by trendy brands, the Earth is paying the price through the process's harmful carbon footprint.

That's where the role of clothing technologists and designers comes in, as their job is to introduce innovative advancements in clothing to make the industry sustainable, durable, and of course, still fashionable.

How can I start?

A number of schools offer a BS in Clothing Technology, such as the SoFA Design Institute, the University of the Philippines Diliman, and the Technological University of the Philippines.



Archaeologist

Yes, there are archaeologists in the country! With the Philippines having a rich culture taking back thousands of years, our own archaeologists have been discovering artifacts that have made the country a vital research ground on human evolution!

Most recently, researchers found two fossil fragments aging almost 50,000 years and 67,000 years in Cagayan's Callao Cave!

How can I start?

University of the Philippines Diliman offers a complete Archaeological Studies Program, where you can attain your bachelor's, master's, and doctorate in the field.



Forensic Scientist

Forensic science, also known as criminalistics, is the role of science during criminal investigations. It isn't as simple as watching crime shows though, as forensics is a broad field that can range from Forensic anthropologists, Digital forensic examiners, Forensic engineers, Forensic pathologists, and Forensic document examiners.

How can I start?

Most forensic science careers can start from any bachelor's degree, though a program in Criminology could get you a leg up on working in the scene. The Philippine College of Criminology in Manila offers a wide array of courses, while there are some universities that also offer the course alone.



Robotics Engineer

The future is now! Being a Robotics Engineer is essentially an interdisciplinary research area between computer science and engineering. The goal of the job is to design intelligent machines that can make human lives easier and safe—more on Sophia the Robot, less 'I, Robot'.

With the growing need for advanced AI in the future, the field of robotics engineering is also on its way up.

How can I start?

Most forensic science careers can start from any bachelor's degree, though robotiq.com says that Electrical engineering is one of the best majors to pursue to help a career in robotics. You can get started through any related course like Mechanical Engineering, Computer science, Mathematics, or Programming.



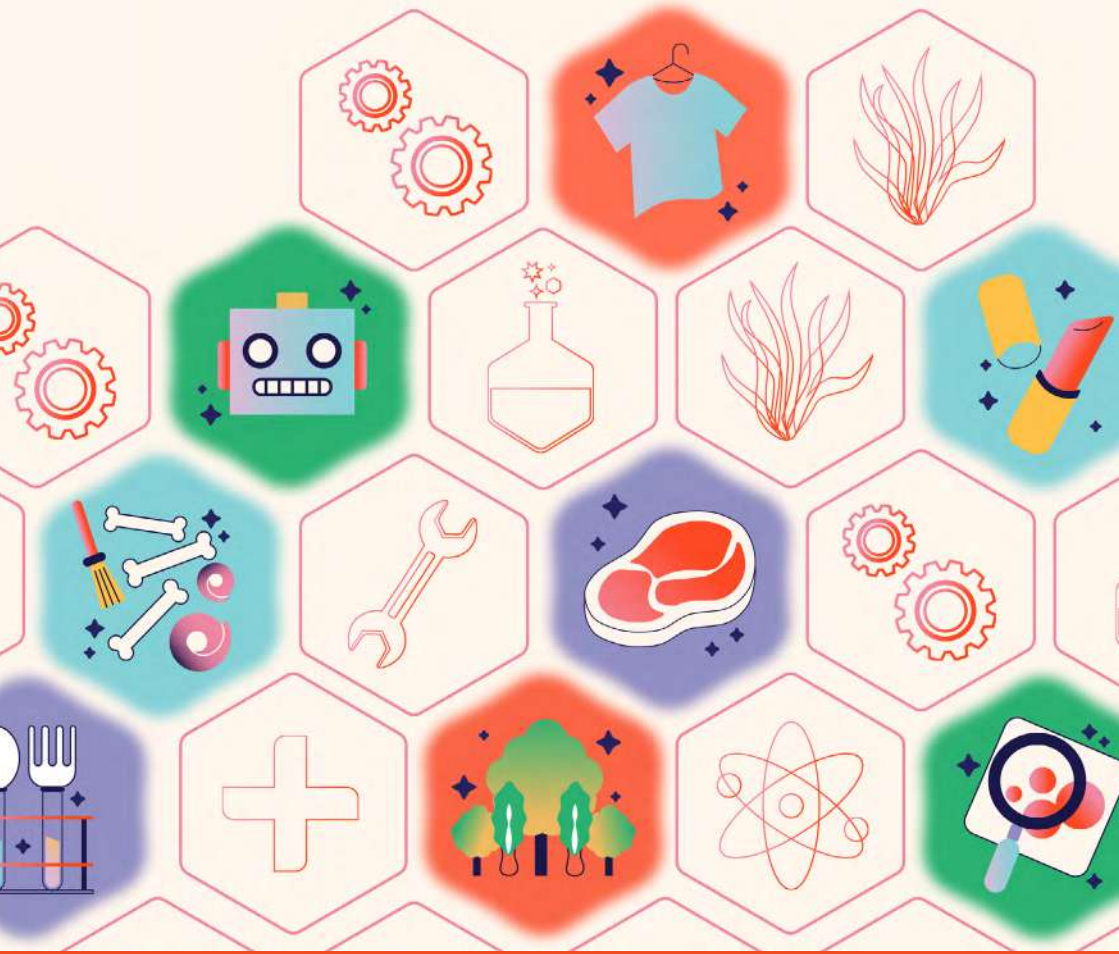
Carnivore Ecologist

Carnivore ecologists study—you guessed it, carnivores! The ecologist part of the job description involves exploring how carnivorous animal and plant species affect each other and their environment. They also research how human-modified landscapes can affect a carnivore's behavior patterns.

One of the most prominent Carnivore Biologists in the world is Dr. Rae Wynn-Grant. She's currently studying the ecological and social drivers of human-carnivore conflict.

How can I start?

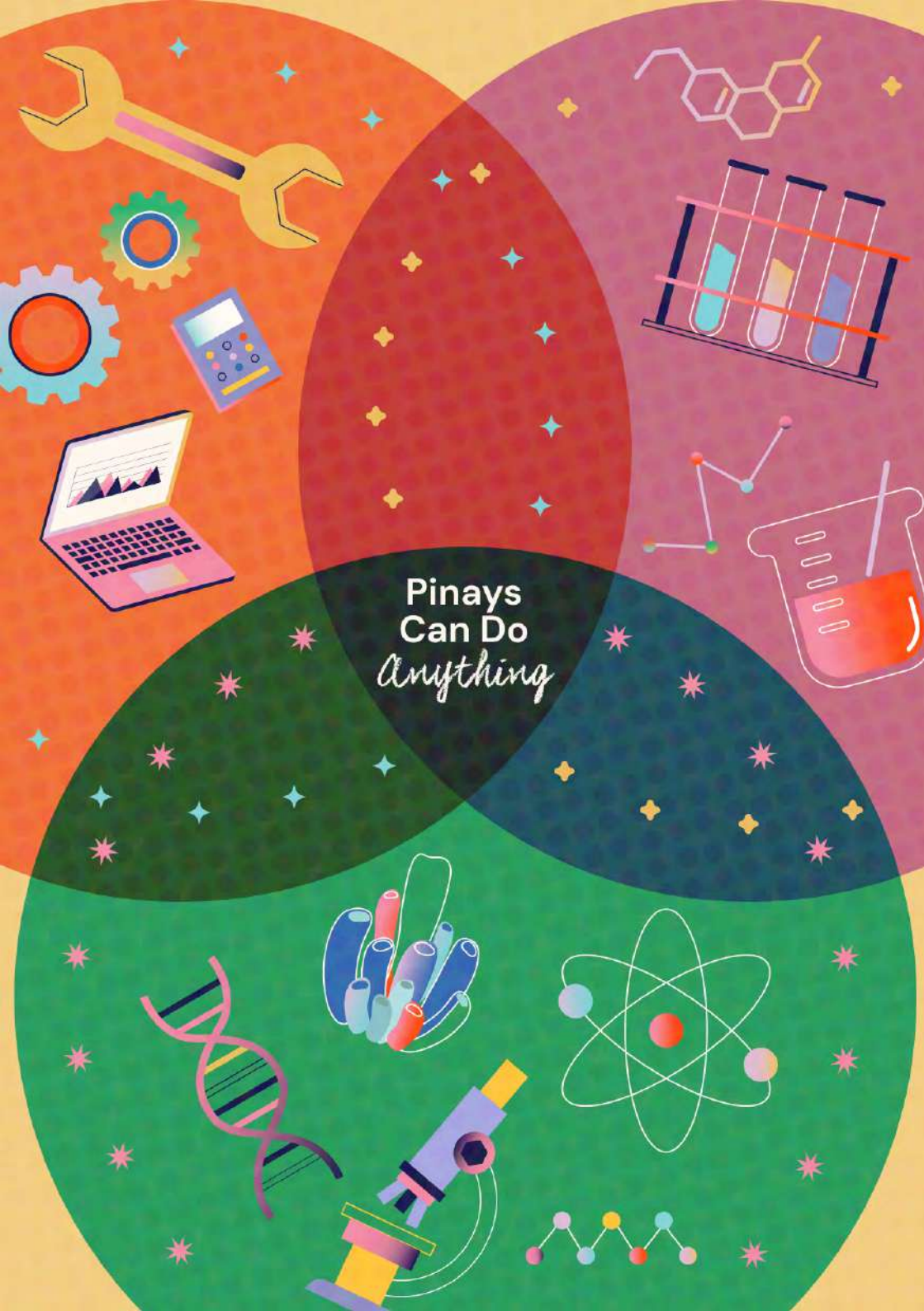
Since the job is still a rare one in the country, it's best to start with learning about Ecology as a whole through STEM-related courses and research.



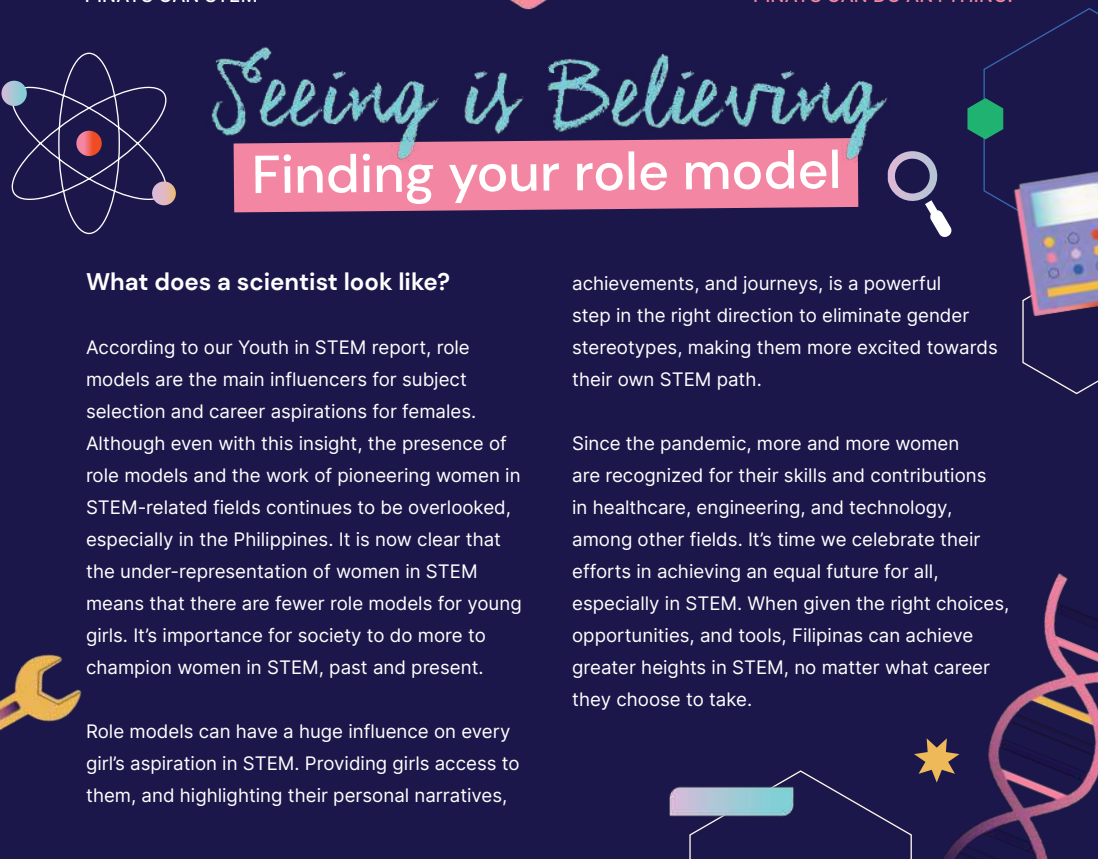
The world moves at a fast pace and so does the demand for more nuanced workers in pretty much any field.

We can't predict what's to come in the next ten or twenty years, but with today's new callings like gamer, vlogger, and streamer bringing in new talents and opening new doors, it's high time for the non-traditional and unusual passions to bloom in STEM too!

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Pinays
Can Do
Anything



Seeing is Believing

Finding your role model

What does a scientist look like?

According to our Youth in STEM report, role models are the main influencers for subject selection and career aspirations for females. Although even with this insight, the presence of role models and the work of pioneering women in STEM-related fields continues to be overlooked, especially in the Philippines. It is now clear that the under-representation of women in STEM means that there are fewer role models for young girls. It's importance for society to do more to champion women in STEM, past and present.

Role models can have a huge influence on every girl's aspiration in STEM. Providing girls access to them, and highlighting their personal narratives,

achievements, and journeys, is a powerful step in the right direction to eliminate gender stereotypes, making them more excited towards their own STEM path.

Since the pandemic, more and more women are recognized for their skills and contributions in healthcare, engineering, and technology, among other fields. It's time we celebrate their efforts in achieving an equal future for all, especially in STEM. When given the right choices, opportunities, and tools, Filipinas can achieve greater heights in STEM, no matter what career they choose to take.

Get to know our STEM Pinays all around the world! No matter where you're from, if you took a nonconventional STEM path, these women are here to tell you that you have a place in the field. Be inspired by their stories, testimonials, and journey in STEM!

Audrey Pe,

The Teen CEO



Audrey (@audreyisabelpe on Instagram) is the CEO and Founder of the nonprofit organization WiTech (Women in Tech) which she started at only 15. The initiative has since impacted over 1,200 young Filipinos through the first WiTech conference back in 2018, tech literacy programs in public schools, a career roadshow on closing the gender gap, blog stories on women in tech role models, and more.

STEM Spark

Audrey got fascinated with tech when her elementary computer teacher strayed from the syllabus and taught them simple coding. This was the first time she learned that websites were made of lines of code—as just a slight tweak from the country's typical curriculum could leave a massive impact in a STEM girl's journey.

How do you envision the future of women in STEM in the Philippines?

I am motivated by a future where all youth, especially in developing countries like the

Philippines, have access to technology and can use it to help solve problems within their own communities. Now more than ever, we are seeing how the digital divide limits students' access to educate in light of school closures. The inability for many schools to not transition online due to not having access to the internet is a barrier for many students reaching their full academic potential. I work towards a future where tech isn't a barrier, but instead a tool to help contribute to one's society and achieve an education that everyone has a right to.

What is your message for our young Pinays in STEM?

At WiTech, we believe that helping eliminate gender inequality and tech inaccessibility work hand-in-hand. To us, equality in the tech industry isn't just 50/50 representation but also ensuring that those in tech represent more than just the upper class. We believe that diversity should come in the form of increased opportunities for all gender and socioeconomic strands so that tech turns into a right instead of a privilege.

SHARE THE BLOG

Hillary Diane Andales,

The Space Explorer



Hillary (hillaron.com) started on the map after she won the Breakthrough Junior Challenge in 2017, a global competition for students to inspire creative thinking about science. She soon founded the regional science camp Science Innovations Bootcamp in the Philippines. Now at 21, Hillary studies physics (with minors in astronomy and philosophy) at MIT and regularly holds talks as a science communicator. She has a personal mission to make students 'excited—instead of intimidated' of STEM.

STEM Spark

Hillary started out when she was just 5 years old after reading an astronomy book that was as tall as she was and completely 'blew her mind.' Hillary became even more fascinated with space through the years, jumping from her dream of becoming an astronomer to a now-budding astrophysicist.

How do you envision the future of women in STEM in the Philippines?

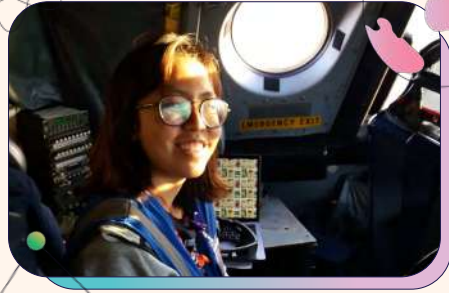
I envision a future where women scientists can just be scientists — period. I hope for a future without a gender gap, a future where it's no longer news when a woman wins a Nobel Prize. For the Philippines specifically, I just hope that the government provides enough support (more research funding, less delay due to bureaucracy, better leadership, and so on) for our Filipino scientists.

What is your message for our young Pinays in STEM?

For young Pinays in STEM: keep dreaming big! Do not let anything limit your dreams. Use the internet and social media to find inspiration, and then take those as fuel for your dreams. Also, as I always say, do not be afraid of failure. When you fail, you don't really fail. You only learn.

SHARE THE BLOG

Bee Leung, The Forecaster



Bee (@thegobidesert on Instagram) is pursuing her Master's degree in Atmospheric Sciences at Colorado State University. Her course uses computer models, field campaigns, and satellites to better understand how clouds are formed. Bee plans to bring her studies to the local scale by discovering how pollution and human-made changes affect the Philippines' humid environment.

STEM Spark

Bee's love for STEM blossomed when she started focusing on improving the lives of the marginalized sector through her work. She first executed projects measuring the exposure of jeepney drivers to air pollution and quantifying how Filipino farmers will become more vulnerable to heat stroke.

How do you envision the future of women in STEM in the Philippines?

I think a lot of it comes down to the details of how we run our organizations and research labs: making sure that people can have enough time to care for their families and other people in their lives, that the workplace and the laboratory are

safe places with accommodations for people's differing needs, that work is being distributed equally and equitably, that everyone is getting the chance to share their ideas and learn from one another.

And of course, I hope that by looking out for one another as we build that kind of positive scientific community, I also hope that Filipina scientists (and all scientists, really) can remember to look out for others and let our compassion fuel the work that we do.

What is your message for our young Pinays in STEM?

I hope you know that you belong here as much as anyone else! Honestly, that's still something that I have to tell myself too, so it's okay to feel insecure or like an impostor. I don't know if that feeling ever goes away, but based on conversations with mentors and other scientists over the years, I don't think it does, and that's okay. If you want to go into STEM and you love what you do, then go for it! Science is for everyone.

You Had Me at 'Molecular Biology': How **Ea Tulin** Fell In Love with Science

Ea Tulin is a third year PhD student in Applied Biological Chemistry at the Tokyo University of Agriculture and Technology. Her research is focused on investigating glycan formation in the central nervous system. She is also a faculty of the Department of Biotechnology in the Visayas State University. Growing up in Leyte province, Ea did not have many opportunities to learn from other Filipino scientists and programs. At a young age, she understood the reality that the media and private companies would invest more

and prioritize Manila-based programs. It was only with her parents' influence and the support from high school and elementary teachers that she grew to love science. This also cultivated her fascination with medical research, biology, diseases, and the brain.

From falling in love with the word 'molecular biology' to eventually pursuing biological chemistry in her doctoral degree, Ea shares with us her journey in STEM.

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Did you know that Ea is also an artist? She recently started a project called **Experiments in Art (EA)** where she showcases mouse brain pictures in creative ways by combining them with art.

What made you decide to pursue science in particular?

Both my parents are scientists. My dad is a biochemist while my mom is a soil scientist. My first experiment was making antibacterial soap with my dad using flowers near my school for an elementary science fair. We won first place. It was definitely a way for me and my dad to bond and my mom was also very supportive, so it wasn't hard to fall in love with the field. Ever since I was a kid, I've always loved working in the lab and doing research. This, along with growing up with the influence of my parents within the (Visayas State) University are the reasons why I decided to pursue STEM.

Why do you think that confidence in STEM is important for our Pinays?

Confidence is important to give us a head start and visibility. I believe a lot of Filipino women are confident, yet they still seem to burn out. I think confidence combined with focus, grit, and kindness, are important and will help us move forward as Pinays in STEM. Collaboration is key, and in a field where everyone is smart, we stand out by having a good attitude.

How can we get Pinay students interested in STEM while at home?

Social media is a good way to attract Pinay students! There are many pages that feature women from all over the world doing STEM: Pinoy Scientists (for Filipinos in STEM), Women Doing Science, Women Transforming Science, 1MWIS (1 million women in STEM), and similarly pages of women scientists talking about their lab lives online. Instagram was a great avenue for me to meet all these women (virtually) while at home!

PINAYS CAN DO ANYTHING!



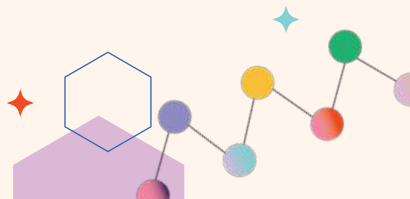
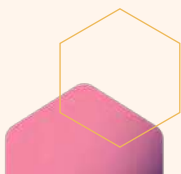
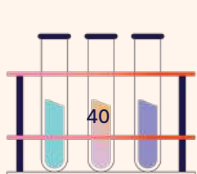
Her first conference in Japan at the 37th Annual Meeting of the Japanese society of carbohydrate research.

"If you feel like STEM is something you would like to pursue, go for it! Find a role model or be your own."

How do you envision the future of women in STEM in the Philippines?

There is a lot of untapped potential in the Philippines and this can be harnessed with proper support from the government, collaboration between universities, and a joint effort between private companies and universities. This is the goal. With movements that push women to be empowered in STEM and Philippines being one of the highest in Asia when it comes to gender equality, I believe women in STEM will continue to rise in number, assume higher institutional positions, and be a key contributor to the realization of this goal.

Diversity broadens the pool of knowledge and expertise in STEM and other related fields. Past generations have worked towards creating platforms and increasing visibility for more women in STEM, and even until today, we strive to break barriers to encourage more young Pinays to pursue STEM tracks and careers.




[SHARE THE BLOG](#)

Let Me Count the Ways: Math in Action with Dr. Ninette De Las Peñas

Math is the language of the universe. From education to mat weaving, it is also how Dr. Ninette came to understand the world.

Most people see math as daunting and complex, a universe of its own irrelevant to the everyday and familiar. However, for Dr. Ninette, numbers are sometimes the best portals to other worlds, especially from our own selves to the greater world around us.

Growing up, Dr. Ninette has always been fond of math. She says, "I [have] loved math since I was a kid and was always intrigued by what I could discover through numbers." From deciding to be a mathematician as early as her high school days until today, she has had a thriving career in her chosen field. Over the years, she has won many research awards, written over 60 journal articles in mathematics and mathematics education, and mentored over 40 thesis students.

Photo taken during a lecture in Ateneo de Manila University back in 2017.



PINAYS CAN STEM

Currently as the Associate Dean for Research and Creative Work of the Loyola Schools, Ateneo de Manila University. Dr. Ninette's most recent project was leading a research team that developed a mobile app for math education, known as Mathplus. The project aims to instill critical thinking skills in children grades one to eleven while also aiding in transitioning to online and blended learning amid the current health crisis.

Aside from being an educator, Dr. De Las Peñas has also used the language of mathematics to communicate other topics of interest as she is best known for her work on woven mat patterns. Alongside two co-authors, she talks about the symmetry of mats from Philippine indigenous groups using the language of mathematics in their work called "Weaving Mat(h)s". The paper was also presented in 2014 in collaboration with mat weaver Janeth Hanapi, showcasing a live demonstration of the weaving process. For their work, the authors received a grant from the National Commission for Culture and the Arts (NCCA).

This is a huge leap from the state of the academe during Dr. De Las Peñas' early professional days. She recalls, "I realized that there were only a few mathematicians from the mathematical community in the Philippines that were published in prestigious and indexed math and science journals," thus her will to develop a culture of publishing as an educator. She also mentions a lack of opportunities to do further research and participate in conferences abroad, among others.



Dr. Ninette De Las Peñas being awarded the 2018 National Academy of Science and Technology Philippines (NAST Phil) Outstanding Scientific Paper Award for her paper entitled "Mathematical and Anthropological Analysis of Northern Luzon Funerary Textile."

In spite of the larger forces that can hinder a STEM Pinay's career, Dr. De Las Peñas still thinks that encouragement and role models for young girls can go a long way, particularly from family and media. She cites Nancy Drew as an example, describing the titular character as "very scientific in the way she found clues to solve her mysteries," as well as assertive and headstrong.

"Do not give up on your dreams. Do not be afraid to assert yourself. If men can do it, we can too!"

She believes that things will only go upwards for women in STEM. "They think differently and have more confidence. There are more opportunities now for women to succeed." As words of wisdom for young girls aspiring to do the same, Dr. De Las Peñas says, "Do not give up on your dreams. Do not be afraid to assert yourself. If men can do it, we can too!"



Overcoming Uncertainties: Quantum Physics with Dr. Jacqui Romero

Ever since her childhood, Dr. Jacqui Romero's interest in STEM was nurtured by an ever-supporting family. "They have nurtured my interest in mathematics and science from a very young age. I remember my father driving me to weekend MTAP lessons!" Eventually, she took a particular interest in physics. Her passion for the discipline started during her high school years in Philippine Science High School. Physics was her favorite subject and she would even read beyond what the curriculum would require. In particular, quantum computing was her gateway into the more daunting field of quantum physics.

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This feat did not come without difficulty. Dr. Romero mentions two most challenging parts to her journey: getting into the field and landing a job. "I found it hard to get into a research program abroad for a PhD [because] there [was] just a lack of awareness of [programs] then." Because of this, she became more vocal about the programs in the field of Physics. She then landed a scholarship to do her PhD, which she accomplished all while pregnant with her first child. While earning her doctorate, she also contributed to eleven publications, five of which were led by her.



Eventually, after completing her post-graduate degree, Dr. Romero encountered her second challenge: the job hunt. She came to realize that entering the field is one thing, but landing a position is another. Due to the scarcity of jobs in the academe, it is quite difficult for each expert in the field to be afforded a position. She mentions, "it is really sad that we have a lot of talented people and not a lot of jobs in the universities." Her way around it was to put herself out there, gaining media attention from her work on slowing down the speed of light in free space and winning fellowships and awards that put herself in the spotlight. To name a few of her accomplishments, Dr. Romero has once been selected as one out of fifteen L'Oréal-UNESCO For Women in Science (FWIS) International Rising Talents. In 2016, she was also awarded a fellowship by the Australian Research Council, then received a fellowship from the Westpac Foundation afterwards. As Dr. Romero proudly exclaims, "I made sure that I cannot be ignored!"

When asked about the future of women in STEM, Dr. Romero hopes that the Philippine government would invest more on research and development. While it is great to have scholarships for science, what follows after a science degree is still unclear without opportunities for research. "It often happens that one finishes a PhD and then pursues a career in administration. This is good, but that is really not what a PhD is for. As abstract as it sounds, we need to generate new knowledge so we can solve more problems. Some of these problems will be unique to the Philippines, and



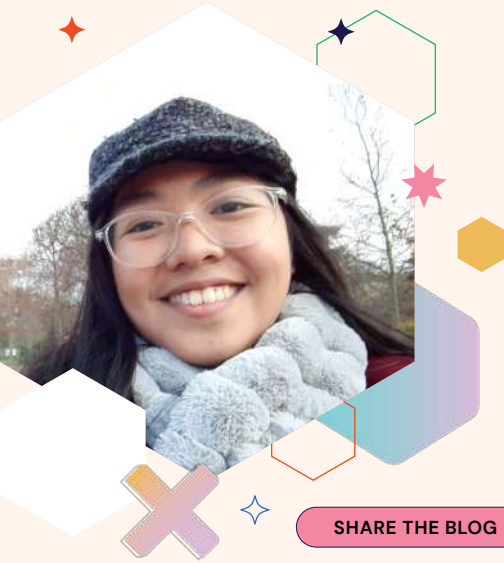
"Be good! Whatever it is you choose to do, be very good at it to the point that you cannot be dismissed. Also, always remember to have fun!"

we would need Filipinos who have the expertise and passion to solve them." Hopeful, she mentions that the Philippines has a wider pool of talented young women based on the number of female students who, as she would observe, would attend Physics conferences each year. It is then a matter of making the most out of this culture through policies and systemic changes.

Dr. Romero's advice to young Pinays looking to venture into STEM is "Be good! Whatever it is you choose to do, be very good at it to the point that you cannot be dismissed. Also, always remember to have fun!"

Dr. Jacqueline Romero is a Reader (Associate Professor), currently working for the Australian Research Council Centre of Excellence for Engineered Quantum Systems at the University of Queensland, Australia. She is also a loving wife and a mother to three children.





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Defying Deviations: Exceeding the Norm with Angelina Aquino

A Pinay in the field of electronics engineering, Angelina talks about her ups and downs as a woman in STEM.

Angelina Aquino has had an admiration for math and the sciences since her childhood. As a young girl, Angelina would join math competitions, and her parents, both engineering graduates were very supportive of her passion. “[My parents]

fostered my curiosity about the world, explained new concepts well, and encouraged me to read books and watch documentaries,” says Angelina. Later on, she found herself in a community of like-minded people, particularly during the math competitions, during her years at a science high school. It was then that she found her love for programming.

Day 1 of the LT4All conference, with former DepEd USec. Dina Ocampo (center) and Prof. Rhandley Cajote (right).



Although she initially thought she would pursue medicine, through her interactions with her teachers, she later realized that another field was more suited for someone as logical and critical-minded as herself. "I soon realized that I couldn't imagine myself working in a field without problem-solving, where you encounter questions and are able to break them down into logical, step-by-step solutions." Eventually, this led her to pursue studies, both a bachelor's and master's degree in engineering.

Her current focus is called natural language processing, or NLP for short. Though we may not realize but NLP is something we use daily. To put it simply, NLP makes use of language data, whether it be written or spoken, and this is often used in the development of apps such as Siri or Alexa, Google Translate, search engines, spam filters, chatbots, speech recognizers—you name it! An interdisciplinary field, NLP pools together the fields of electrical engineering, computer science, linguistics, and many others.

Angelina hopes for this spirit of collaboration for the local science community since she believes this is how new knowledge is formed. "When you come across a question that hasn't been answered before and you start finding new answers—now that's science!" Because of this, there should be more opportunities for women in STEM. While she mentions that the gender gap in the Philippines is relatively smaller compared to other countries, nonetheless, it is a reality that must be overturned for the better. Throughout her years in engineering, she has often heard that this is a man's field, yet she still persevered and succeeded despite society's archaic views on the matter.



Angelina in ISMAC 2019 with her father, who is "arguably the single greatest influence in my pursuit of STEM to date."

"In the future, I envision women to be and feel as welcome in STEM as in any other discipline."

She has nothing but hope for Pinays in the sciences. "In the future, I envision women to be and feel as welcome in STEM as in any other discipline." Her advice to young girls is to live curiously and keep asking questions. "Don't let anyone ever think that you don't have a place in this field. Like this year's Nobel Prize laureates in Chemistry, Emmanuelle Charpentier and Jennifer Doudna, you can be trailblazers in your own way."

Angelina Aquino is currently a Lecturer under the Electrical and Electronics Engineering Institute at University of the Philippines. In her spare time, she enjoys singing and listening to choral music, as well as playing video games.



Healing Together: Erika Modina's Journey with Public Health

A young public health professional, Erika Modina offers insight into her one-of-a-kind STEM journey!

"I want to shatter the notion that you have to fit a certain criterion to pursue a career here, that you can wear head-to-toe pink and still be taken seriously."

These are the words of Erika Modina, a public health professional from the Philippines.

Growing up, she initially wanted to become a doctor because of her proficiency in science, but more so because she did not realize she had other options as well. While she once considered either working for the government or managing a children's hospital, she was enlightened to pursue public health research after she graduated from college with a degree in BS Health Sciences.

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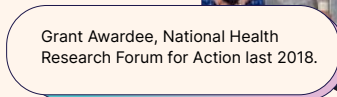
Eventually, she saw to it to enhance her skills and knowledge needed in the field, particularly through further studies. She has since studied in the London School of Hygiene and Tropical Medicine in the University of London and is completing her Masters of Science in Public Health in the said university.

Despite being well-accomplished, Erika has still had her fair share of difficulties. Having completed her bachelor's degree in 2016, she is a relatively fresh face in the industry, which sometimes leads people to believe that she is too young or lacks experience. The fields of public health and medicine are still quite traditional in the Philippines, which gives her the amount of naysayers who doubt her skills. However, like the headstrong woman that she is, she chose to prevail and simply let her work speak for herself.

Prince Mahidol Award Conference in Bangkok, Thailand last 2020.

Erika finds motivation in knowing the stories of others. Instead of self-help books, she looks to memoirs to appreciate what others have been through which, in turn, makes her excited for her own story to unfold. She encourages women to never hide their accomplishments. Given the current health crisis, her advice is to also use this time to make conversation with other people. "This is the perfect time to reach out to other women you look up to or people you want to work with. Take this time to find your tribe and surround yourself with people who make you kinder and better, not just in your career but in all aspects of your life."

Her advice to young Pinays? "If one person belittles you, look around—ten more people will be cheering you on." Each person's STEM journey is unique. For Erika, it is a journey "towards health equity, lined with peonies, [with] 'What Dreams Are Made Of' by Hilary Duff blasting in the background."



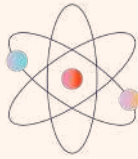
Grant Awardee, National Health Research Forum for Action last 2018.

"If one person belittles you, look around—ten more people will be cheering you on."

Erika Modina is currently the president of EpiMetrics, Inc., a health research institution geared towards the achievement of health equity through rigorous and creative conception, execution, translation, and communication of health systems and policy research. She is also the Chief Health Officer of Day3 Innovations and a part-time lecturer at Ateneo de Manila University.



Erika participated as a panelist in Startup PH's Women's Initiative last 2019.



A Deep Dive into Marine Biology with Jean Utzurum

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Writing, doing field work, and sometimes teaching people how to swim—marine biologists do it all. Hear from Jean, one of our most passionate Filipina marine biologists.

Although already well-accomplished in the field of marine biology, Jean would describe her journey in STEM as never-ending. Having completed her bachelor's and currently undergoing graduate studies, she initially never considered becoming a scientist. For various reasons, Jean felt limited to a life in academia or working for an environmental NGO after completing her bachelor's degree. In graduate school, she saw a world beyond science where research leads to policies and action. It was then she realized that the sciences are far from fruitless.

While passionate in her pursuit of science, Jean admits to occasionally having to meet halfway between career and practicality. "There have been many times when I've had to re-prioritize research goals to favor jobs that pay more.

Sometimes when I'm in between jobs I'll even do freelance work as a swim instructor and writer. It's during these in-between days that I'll find myself reevaluating a career in STEM." Although surrounded by supportive friends and family who are nothing but encouraging, the reality of having to make a living is something she also considers.

In 2013, Jean was part of a coral rehabilitation project in Apo Island, which was established through the island's Protected Areas Management Board (PAMB).



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However, Jean has definitely made some notable strides in her field. She found her niche in the study of elasmobranchs, or sharks and rays as known by many. This eventually allowed her to work with the Silliman University's Marine Laboratory Museum and their extensive collection of chondrichthyan, or cartilaginous fish specimens, and become the co-author of their field guide. Recently this year, she was also invited to virtually participate in the regional International Union for Conservation of Nature (IUCN) Red List Assessment Workshops for 126 species of sharks and rays in Southeast Asia.

"I've been fortunate to have worked with and be mentored by fellow women in STEM throughout my career, and to have lived in a university town like Dumaguete City with world-renowned Filipino women scientists as role models (Dr. Hilconida Calumpung, Dr. Louella Dolar, and my aunt, Dr. Ruth Utzurum among others)."

Jean considers herself lucky because opportunities like these are hard to come by. Beyond gender, there are other factors such as lack of opportunities, as mentioned by Jean. "Gender has not been a hindering factor, to say the least. In my case, the lack of long-term employment opportunities in the field of marine biology has been the main factor." On a more serious note, she does acknowledge the gender

Jean at the Visayas SEA camp talking about sharks and showing their published field guide.

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Her fieldwork involved monitoring catches, taking disc width measurements, sex determination, and dissecting them to check for signs of maturity (if female) and occasionally, checking to see what they've eaten.

gap in STEM. While it might not be as extreme a case here in the Philippines, it is still the reality of today's time. "Sexism and harassment have not hindered my pursuit of STEM. But it exists. It happens. It can be very tiring, stressful, and traumatizing to deal with." Nevertheless, she finds hope in initiatives to make safer spaces for women to blossom in the field.

Because of this, she is optimistic about the future of women in the sciences. "Visible, loud, and accepted." This is how she envisions women in STEM. "You can be anything you work to achieve. It's okay to have days when you doubt yourself. Don't be afraid to ask questions —keep being inquisitive and curious!"

Jean Utzurum is a marine biologist who is finishing her master's degree in Silliman University where she also completed her undergraduate studies. While she currently works freelance, her previous field research experiences include coral reef restoration, coral reef and mesophotic fishes, and fisheries. She has volunteered with Reef Check Philippines and the World Wildlife Fund and also served as an elasmobranch specialist for the Marine Wildlife Watch of the Philippines.



Arti-facts Only! Digging Deep into Archaeology with Dawn Satumbaga

Listen to Dawn Satumbaga as she unearths her stories as an archaeologist, wiping the dust off of myths about women in STEM!

Doctors, scientists, environmentalists—it seems that everyone in the sciences has it all figured out. For Dawn Satumbaga, however, her career has admittedly had somewhat of a late start. “Some people already know what they want early in life and plan it out carefully until they specialize—that’s not me,” says Dawn, a Pinay in the field of archaeology. “I went through a math course in college, pursued archaeology for my Masters, and I am currently taking Environmental Science for my Ph.D., so it’s okay to start a bit

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late! But what has always been constant in all this is the drive to learn new things.”

During her undergraduate years, while Dawn majored in math, she also took a minor in English literature. Her curiosity has also led her to other places both local and international. During her masters, she had been invited to join the prehSEA project in Palawan, Philippines, which eventually gave her the chance to do similar work abroad. With funding from the French Embassy, she underwent training and attended lectures and seminars at the *Muséum national d'Histoire naturelle* in Paris, France. Throughout her stint as an archaeologist, her favorite field activity is surveying for sites because it takes her to new places, yet her favorite site is back home in the Philippines, specifically in Callao Cave, Cagayan. “It’s because of the beautiful river we get to swim in every day, and it’s also where I did my thesis!”

Dawn analyzing soil samples in the lab. Taken back in 2015.



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Aside from being motivated by curiosity, Dawn also gives credit to her supportive upbringing. Growing up, she had an encyclopedia set which spurred her hunger for learning. While living in the province proved to have its downsides, she also mentions that living close to nature made her more keen to the phenomena around her. Her family has also been nothing but supportive throughout shifts in her career, while the Archaeological Studies Program community in the University of the Philippines Diliman was also greatly accommodating.

Dawn reflects on the journey she has had with STEM so far. "When I think about it, I might not have made these leaps and planned out my education early on if I had better exposure to different STEM fields and career options." Although she is lucky to have explored various fields and gone where her curiosity has led her, she also mentions some factors which may make it a bit harder for people like her to find a steady path in the sciences.

She brings up the lack of career counseling for scientifically inclined youths, particularly in the provinces. "I'm not sure what the situation is like in Metro Manila, but in the province where I grew up, there were hardly any career talks or fairs for high school students to explore and make more informed decisions." Because the sciences are so diverse and the many things under the sun, one may not have the proper career counseling, making it difficult for people to know their options.

The second factor she mentions is the systemic changes that have yet to come, particularly for women in the field. Archaeology is often associated with treasure hunting or wild adventures to exotic and cursed spaces, but it is a discipline that involves rigorous cultural studies and the hard sciences to make sense of our human past. While archaeology is popularized



Dawn backfilling in Aklan back in 2014.

"Curiosity is the core of science, and this is what makes it exciting, but seeing it work and help others is what makes it worthwhile."

through thrill-seeking males, real archaeology has a diversity of women both in the field and in the laboratories. While it is easy to envision a future where women are celebrated in STEM, Dawn says that "there are still many things that we need to work on, like normalizing breastfeeding, work-from-home arrangements to allow mothers to spend time with their children, creating child-friendly workplaces, and understanding and creating safe spaces to talk about women's health issues that could affect work like PCOS (polycystic ovary syndrome) and dysmenorrhea." Indistinct yet common to women in the field, there is much to work on in both discussing and addressing gender-based harassment. If systems were more enabling and understanding, then Dawn bravely claims that women can rise to their fullest potential.

Dawn also calls on Pinays who want to follow in her footsteps. "In asking for equality, fairness, and respect, don't forget to give the same to others. Good science is a team effort so it's important to help others and ask for help when you need it. You have an important role, but so do others, so have faith in yourself, but don't lose faith in other people, because Pinays can STEM just as much as anyone can!"

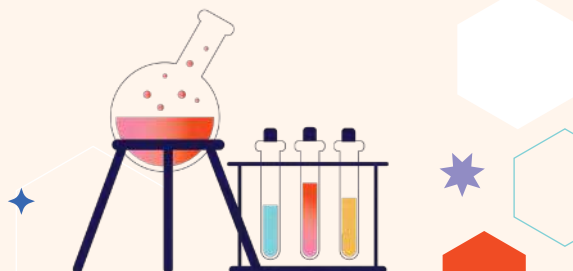
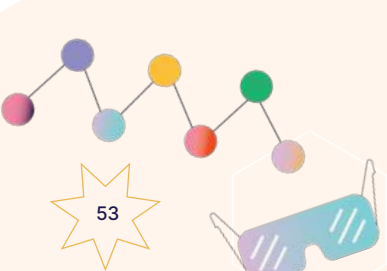
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Chemis-trying to Find Your Element with Herdeline Ann Ardoña

As the saying goes, if you're not part of the solution, you're part of the precipitate! Learn how to make STEM a better place for women from materials chemist and professor, Herdeline Ann Ardoña.

"Developing biomaterials with optical and electronic functionalities that are compatible with excitable cells such as cardiomyocytes"—does this sound like a mouthful? For materials chemist Herdeline Ann Ardoña, there's more to science than the textbooks and the big, highfalutin words. To her, the sciences are where different worlds collide and collaborate to make things better for others. (Speaking of which, those long words are what her team currently does! More on that later.)

Herdeline's STEM journey began with her love of Chemistry. She mentions that while it was unusual to imagine a young girl as a scientist, she took a leap of faith despite it not being the popular choice for people like her. She recalls, "I liked how Chemistry has so many branches—each of them is very different from one another, but all of them are towards understanding the fundamental properties and reactivity of matter. That is remarkably interesting to me." She mentions that her interest grew while she was doing hands-on research. She learned so much from this—from organic chemistry to its applications in biomaterials engineering, this has trained her to take a more transdisciplinary approach in her lab work, as well as solving questions in science and engineering.



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The sciences are so diverse that even within one field of study, there are still so many smaller branches and interests under it. This collaborative energy between and within fields is one of the things Herdeline loves the most about the world of STEM. "In graduate school, I was incredibly happy to be in an environment where interdisciplinary research is fostered." Up to this day, she still carries the same team spirit in her current work. Can you remember the first words of this article? To make those a bit more digestible, her team's task is to control cell or tissue behavior—a crossover between chemistry, engineering, and biology!

Taking a look at Herdeline's upbringing, it's no surprise how she ended up studying chemistry. If systems were more enabling and understanding, then Dawn bravely claims that women can rise to their fullest potential. She says, "It had so much impact on me to see, from a young age, a woman working in STEM." "Because of this, she never saw her gender as a hindrance to her work, yet some other factors made it a bit harder for her to get to where she is today.

"One of the hindering factors for Filipino researchers is the limited availability of funding to support STEM research, as well as the facilities and equipment required to conduct research." For this reason, Filipino scientists often look elsewhere and seek opportunities abroad to get proper hands-on training. Herdeline also mentions that there are so many skilled and talented scientists and researchers in the Philippines who are less fortunate to have the same resources. She expresses her hopes by saying, "This experience is something that equally talented undergraduates back home do not necessarily get. I wish that this situation will change eventually."

However, not all hope is lost. She also gives credit to online avenues (and even gives Pinays

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"Always remember that those successes will not only serve you but can also open the doors for the next generation of girls behind you."

Can STEM a little shoutout!!) in inspiring aspiring Filipino scientists to take on the path to succeeding. "Platforms such as Pinays Can STEM should be continuously supported so that young Pinays can learn about the different trajectories that one can take to be successful in a STEM career."

Knowing the stories of others can truly give us a better look at what options we have, but Herdeline also conversely says that our journeys could help others make sense of theirs as well. "There is no single, best path for girls who want to pursue a career in STEM. It is up to you to find and follow the path where you'll be most excited. This is not an easy career path, but it is your enthusiasm towards small steps that will lift you towards bigger successes. Always remember that those successes will not only serve you but can also open the doors for the next generation of girls behind you."

Herdeline Ann Ardoña is a materials chemist hailing from Valenzuela City, Philippines. She completed her BS in Chemistry at the University of the Philippines, Diliman and received her PhD in Chemistry at Johns Hopkins University, funded by Schlumberger Foundation and Howard Hughes Medical Institute. For 3 years, she was an ACS Irving Sigal Postdoctoral Fellow at Harvard University. Currently, she is an Assistant Professor at the University of California, Irvine Department of Chemical and Biomolecular Engineering.

Biology and Theatre?! The Crossover We Need with Gracetine Magpantay

Crossing over doesn't only happen in our chromosomes. Lecturer of Genetics Gracetine Magpantay recounts her journey in the sciences—from selecting a course in university because of a Taiwanovela to her teaching career today, with a few detours into the arts along the way!

One thing she was certain of was that she was good at math. The rest, although riddled with uncertainty, led her to her career today.

Like some commitments in life, people may join for the wrong reasons but stay for the right ones. While there's nothing exactly wrong with choosing a path because of a television series, it is quite an unusual beginning to a long-term

pursuit, much like that of Gracetine Magpantay, a biologist specializing in the field of genetics. She recalls, "Because of a Taiwanovela, I chose Biology when I took the UPCAT. I passed not knowing what the course is really about." Although a seemingly funny anecdote, this is evidence of a lack of proper career orientations for high school students in the country, especially those who reside outside of Metro Manila.

Candidly, Gracetine admits to other times she half-heartedly trudged on. "I also did not like Biology in [high school] because it was not taught properly. I was planning to shift to either BS Mathematics or BA Communication Arts when I was in [my second year of university], but I did not want my parents' money to go to waste, so I pushed through." Gracetine eventually studied genetics and fortunately grew to have a passion for it. Having always been intrigued by life's mysteries, Gracetine says that studying the building blocks of life made her want to know even more.



Feeding her curiosity, Gracetine made a career out of the Biology degree she at first reluctantly chose. Following her undergraduate degree, she pursued further studies and completed her MS in Genetics, Genetics cognate in Molecular Biology, and Biotechnology. She has also contributed to the International Rice Research Institute's gene editing team and at their C4 Rice Center.

An advocate for women's representation in the sciences, Gracetine also cannot deny the pull arts has on people's decisions. After all, it was a show that got her into Biology in the first place, but beyond the level of fascination, shows can also be used to empower women. As Gracetine puts it, to encourage women to get into STEM, we must "publicize success stories of women in science and create shows about these stories" She also hopes to participate in the production of these shows.

No stranger to this, Gracetine also has equally impressive credentials in the field of theater. With years of experience under her belt as an actor and occasionally as a stage manager, costume designer, and makeup artist, she is what one might consider a Renaissance Woman. Currently, she is balancing studying Theater Studies with her job teaching Genetics.

Gracetine has also found a way for her passion in the sciences and the arts to intersect. One work co-written by her is *Agra*, a musical which tackles issues on Philippine agriculture. Centered on the fictitious *Siporia*, an enhanced rice species, and with a strong female protagonist named *Agra*, the ethno-fantasy play makes it clear that agriculture is inseparable from the greater society.

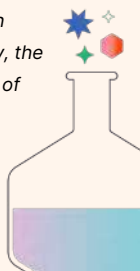


"We must empower each other and believe in each other's capabilities. We must build each other up."

In general, she has high hopes for interdisciplinary projects such as *Agra*. "My dream is to remarry science and arts as how polymaths did in the Renaissance period. Times are hard, and this is the time to join forces and empower STEAM," referring to the integration of Arts into STEM. While not everyone can write a play and teach Genetics, anyone can enter the field of STEM. As Gracetine puts it, "We must empower each other and believe in each other's capabilities. We must build each other up."

Her advice to girls who doubt themselves is simple: "You can!"

Gracetine Magpantay teaches Genetics at Lyceum of the Philippines University in Laguna. She is also the Secretary General for Young Professionals for Agricultural Development (YPARD). She is pursuing her Masters of Arts in Theatre Arts (Theatre Studies) in University of the Philippines Diliman. Currently, she is an Intern at Tanghalang Pilipino Actors Company, the resident theatre group of the Cultural Center of the Philippines (CCP).




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Love the Way You Outlie-r: Beating the Odds with Erika Legara

Physics-major-turned-data-scientist Erika Legara shares with us her STEM journey, from her humble beginnings as a scholar to her current job as a data scientist.

"My journey in Science started as a fortunate stroke of serendipity."

Although Erika wanted to be a scientist when she was little, her run of luck began when she was offered a scholarship by the Department of Science and Technology (DOST). A recipient of the DOST-merit scholarship, Erika Legara originally wanted to take on a different career growing up. "I had always wanted to be an engineer because both my parents are civil engineers," she recalls, but due to the limited course offerings covered by the scholarship, Erika chose Physics since it was the closest science to civil engineering.

Even then, choosing a science course is totally different from staying in it. Erika reveals what keeps her committed to the field of Physics.

"What made me really want to continue pursuing the field were my research mentors," referring to her instructors at the National Institute of Physics and her days at the Instrumentation Physics Laboratory. It was then she found joy in research and collaboration.

Currently a data scientist, Erika's days as a Physics student led her to her pursuit of knowledge, specifically writing simulations and algorithms to explore some what-ifs of day-to-day living. In particular, she has quite a number of interesting research projects under her belt. Not a fan of traffic? Data can solve that! Erika has done research on cities and transportation systems to find ways to make them smarter, more efficient, and more reliable. Bothered by the trolls that plague the internet? Data can offer some perspective on that, too! She has also done some work in Computational Social Science, describing how bots and trolls behave online. "With the right information, the right lens, and the right tools, I, together with our research team, get to help enterprises make better decisions that improve both their business processes and products."

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While a successful, self-made Pinay in science, Erika would not be where she is today without a little help. “The four biggest contributing factors in my pursuit of STEM are my parents, the DOST, my research mentors, and the field of Science itself.” The first exposed her to career options related to civil engineering, the second opened doors, and the third kept her committed. But at the end of the day, it’s science itself that made her stay. “Even if we have the most inspiring parents and mentors, if the field of Science were not as interesting and as mind-blowing as it is, I really wouldn’t have stayed in STEM and continued this pursuit. There’s this deep sense of fulfillment in discovering and creating things, and this is what keeps me in the field and in my profession as a scientist and a professor.”

As for young Pinays who would want to explore the same path, Erika mentions that the road doesn’t come without any challenges. In particular, she mentions that the country doesn’t have a deep appreciation for the sciences, which is reflected in how little scientists and researchers get paid, relative to the years of

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studying and investment. That being said, while outside factors may be to blame, Pinays themselves have the guts to pursue their dreams. With little exposure, aspiring scientists can go a long way! “I am not that worried about building the confidence of our young Pinays. We just really need to expose them to the wonders of STEM.” Nowadays, in the midst of a health crisis, there are still opportunities to be found indoors, such as online internships and webinars which Erika recommends. Some examples include gathering scientists to volunteer or take part in mentoring activities and project-based programs where they can guide young girls in building AI models to perform classification tasks or teach them how to write cellular-automata models in order for them to learn more about segregation and/or land-use design, or to figure out how to best represent social interactions through complex networks. With all this excitement and enthusiasm, she hopes to bring STEM closer to the public, especially to the Filipino youth.

“I am not that worried about building the confidence of our young Pinays. We just really need to expose them to the wonders of STEM.”

Erika engaging business leaders and policymakers. Here, she discussed how Science and Technology can help enterprises survive, thrive, and be competitive in the Fourth Industrial Revolution.

Her final motivational words for us are about STEM Pinays as more than just inspirational figures. “We are aware that as women in science, we have the responsibility to inspire more women to get into science. But more than information disseminators, we are also very much capable of becoming discoverers and generators of ideas and knowledge—something that I would really love to see more of [and] just keep on pushing and persevering. STEM is gender-neutral. Keep on learning, exploring, and creating!”

Knowing the stories of others can truly give us a better look at what options we have, but Herdeline also conversely says that our journeys could help others make sense of theirs as well. “There is no single, best path for girls who want to pursue a career in STEM. It is up to you to find and follow the path where you’ll be most excited. This is not an easy career path, but it is your enthusiasm towards small steps that will lift you towards bigger successes. Always remember that those successes will not only serve you but can also open the doors for the next generation of girls behind you.”

Erika Legara is a data scientist who completed her bachelor's, master's, and postgraduate degrees in Physics under the University of the Philippines. Currently, she is the Associate Professor, Aboitiz Chair in Data Science, and the Program Director of MSc Data Science in the Asian Institute of Management.



At the Asian Forum on Enterprise with women leaders in Science and Technology (AI and Robotics), along with the rest of the panel, Erika explores the central question, “if we can re-imagine, shape, and inspire the Future together, what will it look like?”

Erika in a research meeting, building models to simulate how to better build urban systems and make them more resilient to natural hazards.



Erika training the next generation of data science leaders.

It Really Hertz! The Amps and Downs of Electrical Engineering with Elgelyn Bardelosa



SHARE THE BLOG

Know more about life as a woman in Electrical Engineering from Elgelyn Bardelosa, from her days as a scholar to her supercharged career in Electrical Design Engineering!

It is said that one opportunity can open a lot of other doors, yet sometimes, exploring them can be quite a maze! For our featured Pinay, her journey began as an aspiring high school student, hoping for a place in the sciences. Coming from a science high school, Elgelyn Bardelosa took the admission test to enter into a Civil Engineering course, yet finances were short and she almost lost hope. "Sadly, even if I begged my mother for money, my family's resources were very limited to even afford the fare to go to UP Los Banos. That

day I cried hard because I lost hope—I would not be able to go to my dream school, or even maybe pursue college". Luckily for her, days later she received a scholarship from the Department of Science and Technology, Science Education Institute (DOST-SEI) after passing the exam.

Although Elgelyn did not pursue the course she initially wanted due to the limited course offerings by the DOST-SEI, she went on to pursue Electrical Engineering, and has been committed to the field ever since. Elgelyn is an Electrical Design Engineer by profession. "I prepare electrical plans for different establishments—from a simple house to commercial living spaces like apartments, to small offices to towers, recreational spaces to malls, and even for industrial plants." Throughout her career, she has learned to work in different contexts, sometimes preparing plans for buildings in other countries, and work as a team player.



Although skilled in her field, her journey did not come without setbacks. Initially, not a lot of people were accepting of her course, including her parents who thought she would have had a better career had she chosen Accounting. On that note, she has faced some other challenges that came with her being a girl in the sciences. While her male peers and classmates were welcoming of her, it was during her commutes home after late-night groupworks when she felt unsafe. "Most of the time, I ask one of my guy classmates to send me home early!" To this day, she still has her fair share of discrimination. During her stint as the youngest in the project team, it was difficult for her teammates to take her seriously, and she was also sometimes bullied on the project site for being a young, skinny woman. Resilient, Elgelyn persevered. She bravely states, "I did not let these chances define me as a person, and I took them as chances to strengthen my character. I chose to nurture my technical and people skills to be able to handle those kinds of situations."

Even though these tough experiences can build character, one doesn't need to put themselves in these situations to give them strength. Instead, Elgelyn suggests that young Pinays look for opportunities empowering content online. "It is a very interesting time to pursue STEM in general," she says due to limitations imposed by the community quarantine. While it is hard to spark physical teamwork and collaboration, something ever-present in the field of Electrical Engineering, the internet is still a powerful tool for experts and learners alike, making resources available, even from mentors who are out of the country. In fact, during the quarantine, Elgelyn was able to attend many seminars available online to enhance her engineering knowledge. Aside from



"As I involve myself in many projects and different fields of engineering, it makes me proud to encounter other Pinay Engineers, Lead Engineers, Project Managers, and other Pinay Leaders who also continue to dominate the engineering field."

educational content, she also mentions looking for motivational words online. "I personally go to TEDx talks of women in STEM on YouTube for a source of inspiration."

Elgelyn also offers her own words of wisdom to young aspiring Pinays. "The reason why the STEM field is dominated by men is because mostly known scientists and innovators known before are men. However, we live now in a world that offers a better opportunity for us young women to shine. Choose what battles to fight and what not to. If you don't believe in yourself, remember that I believe in you!"

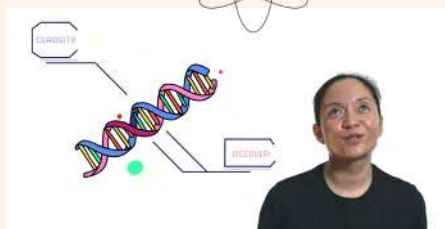
Elgelyn Bardelosa is an Electric Design Engineer based in Cavite, and Electrical Design Engineer at WSP Manila Inc. She graduated from the Technological University of the Philippines-Manila with a degree in BS Electrical Engineering, with a scholarship granted by DOST-SEI.

Pinays Can STEM

TV

Welcome to our channel! Get up close and personal with our Pinays in STEM as they talk about their work, journey, as well as the motivation and inspiration to power on!

PCS TV CHANNEL



WATCH HERE

#MySTEMWork: Understanding the Basics of How Science Works with Dr. Inez Ponce de Leon

Dr. Inez Ponce de Leon is an Associate Professor in the Department of Communication from the Ateneo de Manila University. In this video she talks about the importance of understanding science—how it's always ongoing and ever changing, and how it's more about asking questions rather than finding answers.



WATCH HERE

#MySTEMJourney: Meet Dr. Reina Reyes, the Pinay who proved Einstein right

Dr. Reina Reyes is an astrophysicist and data scientist. She is known for being the "Filipina who proved Einstein right" after her work confirming his Theory of Relativity in a cosmic scale last 2010 during her PhD studies in the United States. She received an achievement award for her research on supermassive blackholes called quasars. In this video, she talks about how her journey in STEM and the importance of being an advocate in the field!



WATCH HERE



WATCH HERE



WATCH HERE

#MySTEMRoleModel: Jammy meets Angelyn, a University Researcher

Our super gals Angelyn Mercado, a University Researcher for Underwater Robotics, and STEM student Jammy, share their experiences in STEM and the importance of providing more opportunities for Filipinos interested in robotics.

#MySTEMJourney: Dive in with Dr. Aletta Yñiguez and discover the beauty of marine science

Dr. Aletta Yñiguez is a Marine Biologist at the UP Marine Science Institute. She led the development of the first integrated biophysical models for harmful algal-bloom (HABs) management in the Philippines. HABs, or more popularly known as red tide, refers to the accumulation of harmful phytoplankton in a large body of water. She believes in working with communities in order to ensure sustainable fisheries in the country and preserve our marine biodiversity.

#MySTEMWork: What You Need to Know About Food Safety with Dr. Pierangeli Vital

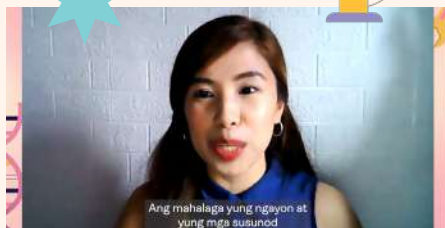
Dr. Pierangeli Vital is an Environmental microbiologist, and Head of the Biological Research and Services Laboratory at the Natural Sciences Research Institute in the University of the Philippines Diliman. She talks to us about the importance of food safety, the careers you can enter, as well as her research work, which focuses on answering food safety issues and identifying contamination in agricultural products and sources.



WATCH HERE

#MySTEMRoleModel: Kat meets Chona, a Molecular Biologist

Meet our Pinays, STEM student Katrina Tee and Chona Abeledo, a Molecular Biologist, and professor at De La Salle University. Chona develops technologies to aid farmers and fishermen maximize their yield and improve their use of resources. Watch as they share their interest for STEM and the environment, as well as the importance of communicating science to the public!



WATCH HERE

#MySTEMJourney: The Vet Is In! Meet Dr. Wenchie Lumbea, Veterinarian and Assistant Professor

Dr. Wenchie Lumbea is a licensed Veterinarian and Assistant Professor at University of the Philippines – Los Banos, where she teaches general, systemic, and clinical pathology for undergraduate DVM students. In this video, Dr. Wenchie shares how it's her drive to learn and remain curious, as well as support from her mother, that made her choose STEM. The ability to connect how STEM can be applied in everyday life is her ultimate motivation to do her best to help everyone around her.



WATCH HERE

#MySTEMRoleModel: Dona meets Ria, a Game Developer

In this video, Ria and Dona share their love for STEM and how they can use their skills and abilities to help change the world.

Ria Lu is currently the Executive Director of Game Developers Association of the Philippines and Founder of Komikasi Games and Entertainment. Dona Marabe is a Grade 12 STEM student and a member of youth-led nonprofit organization Women in Tech (WiTech).



WATCH HERE

#MySTEMWork: The Wonders of 3D Printing with Aisa Mijeno

Are you interested to learn more about how 3D printing works? Catch Aisa Mijeno as she explains the wonders of 3D printing, its applications, and even a demonstration of how to operate it.

Aisa is currently the co-founder & CEO of SALT (Sustainable Alternative Lighting), a social enterprise that develops LED lamps that run on just table salt and water. She is also currently an Engineering faculty member from De La Salle Lipa, head of the Nexus Fabrication Laboratory, and the Deputy Director of Nexus Innovations Lab.



WATCH HERE

Pinays in Digital: Miz Insigne

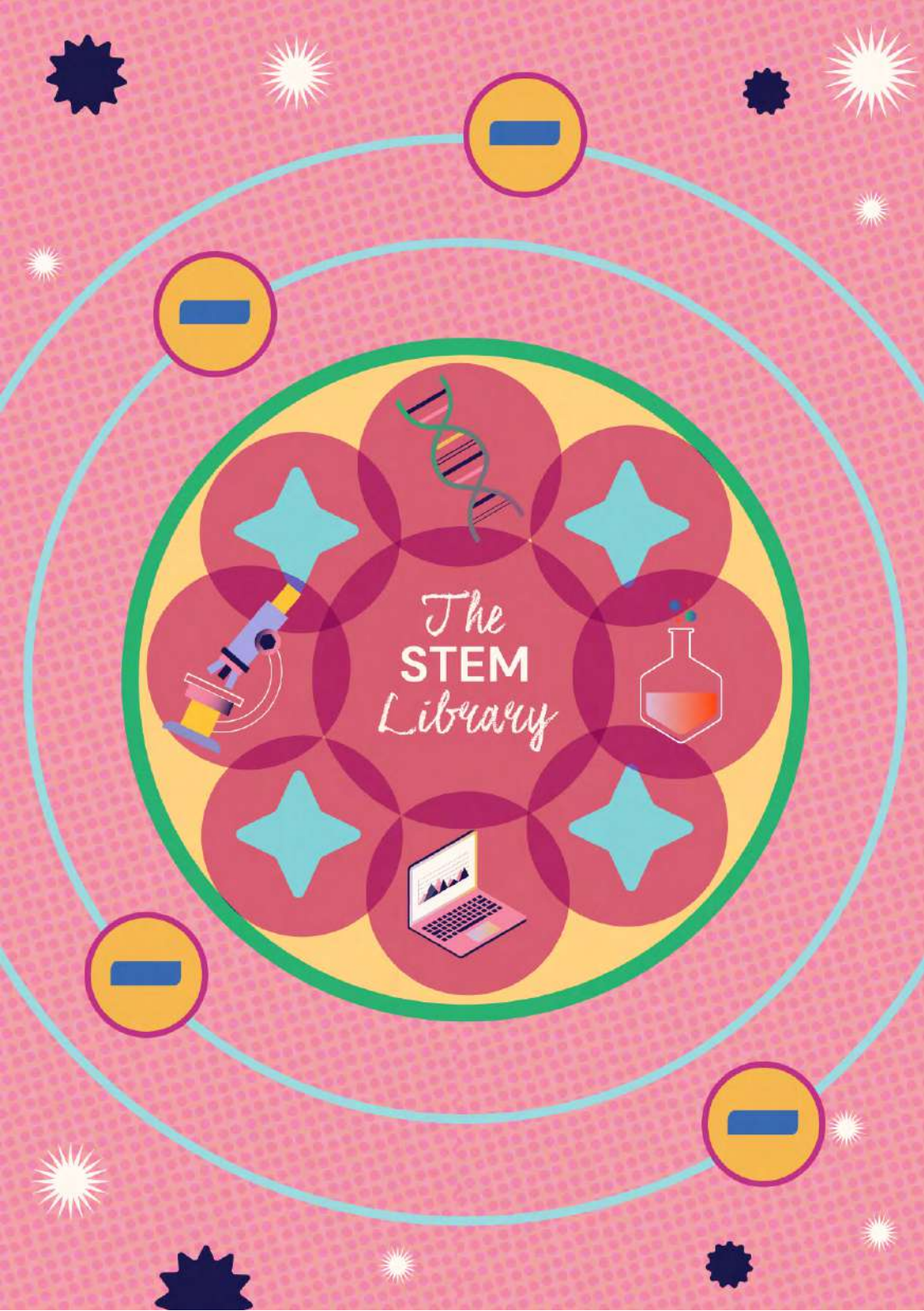
Meet Miz Insigne, currently the co-founder and co-head of SciCreate, an online magazine that promotes STEM and the humanities. As an advocate for STEAM (Science, Technology, Engineering, Arts, Mathematics), watch as she talks more about her growth working in digital spaces and how her organizations have helped shaped her interest in STEAM!



WATCH HERE

Pinays in Digital: Chloe Abrasada

Chloe Abrasada is currently an Occupational Therapy student. She is an advocate for mental health and environmental sustainability, and volunteers in different NGOs to empower and help communities. Chloe has also worked on various passion projects related to her causes, where she hopes to bridge gaps through creation, education, and community mobilization through her work. Listen to Chloe as she talks about using STEM and your passions to create positive impact in the world.



The
STEM
Library



STEM isn't just about **what's in the lab**. There are many resources for you to explore to further grow in STEM! Check out these fun podcasts, movies, documentaries, activities, and other learning resources handpicked for girls in STEM.

[Learning Resources](#)[Podcasts](#)[Videos and Documentaries](#)[Reading resources](#)[STEM Experiments](#)[Templates and wallpapers](#)

Learning Resources

TESDA Online Program

[ACCESS NOW](#)

TESDA Online Program (TOP) is a web-based platform that offers free Massive Open Online Courses (MOOCs) for the technical education and skills development of Filipino workers.

Codecademy

[ACCESS NOW](#)

Codecademy is an online interactive platform that offers free coding classes in 12 different programming languages, including Python, Java, Go, JavaScript, Ruby, SQL, C++, C#, Swift, and Sass, as well as markup languages like HTML and CSS.

PinoyScience

[ACCESS NOW](#)

Pinoy Science is the social media campaign of DOST-PCIEERD, the Innovation Council, which aims to promote learning science and technology to the general public. Follow their TikTok account (@pinoyscience) for more entertaining STEM trivias!

Office Excel Friend

[ACCESS NOW](#)

Want to enhance your Excel skills? Check out this account for more tips, tricks, and content on how you can work better with Excel!

Learn Math and English with Coach Lyqa

[ACCESS NOW](#)

Coach Lyqa was a Civil Service Exam toptotcher in 2013. Since the pandemic, she has released more educational videos on her TikTok account that explains Math and English concepts in minutes! She explains how to quickly solve fractions and equalities, factor binomials and trinomials, and even comprehend word problems. You may also follow her YouTube channel (@teamlyqa) for more content!

CodeGrit Jr PH

[ACCESS NOW](#)

CodeGrit Jr is a coding academy for kids offering one-on-one coding classes with Filipino instructors online.

Science Kwela

[ACCESS NOW](#)

Science Kwela, moderated by Teacher Immanuel Maglasang, is an online channel that helps Filipino students learn STEM in a fun and easy way! Follow their TikTok account (@science.kwela) for more.

Nancy Bullard

[ACCESS NOW](#)

Nancy Bullard is a science-lab teacher at Huntingtowne Farms Elementary. Follow her on TikTok and Instagram (@mrs.b.tv) to watch fun and interactive science lessons for her students while in remote learning.

Podcasts



#STEMTalks

[LISTEN NOW](#)

#STEMTalks is a podcast that sheds light on topics that revolve around STEM—about the field, the misconceptions, gender diversity, and the future of STEM education and the workforce. Together with STEM experts and professionals, #STEMTalks hopes to inspire Filipinos to work together to #STEMify the Philippines!



FlipScience: Ask Theory

[LISTEN NOW](#)

Hosted by FlipScience journalist Mikael Angelo Francisco, the Ask Theory Podcast shines the spotlight on Pinoy scientific brilliance. Each episode of Ask Theory features a Pinoy scientist from one of the many different scientific disciplines.



Scope Podcast

[LISTEN NOW](#)

Scope Podcast is dedicated to topics about science, technology, innovation, and business through the voices of women directly involved in those fields. They discuss current events, interview experts, and engage with listeners through book clubs, movie chats, and question and answer segments.



Superwomen in Science

[LISTEN NOW](#)

Scope Podcast is dedicated to topics about science, technology, innovation, and business through the voices of women directly involved in those fields. They discuss current events, interview experts, and engage with listeners through book clubs, movie chats, and question and answer segments.



The Naked Scientist

[LISTEN NOW](#)

The Naked Scientists flagship science show brings you a lighthearted look at the latest scientific breakthroughs, interviews with the world's top scientists, answers to your science questions, and science experiments to try at home.



Girls Talk Math

[LISTEN NOW](#)

Girls Talk Math is a 2-week summer program with the goal to promote women in math. Girls work in groups to understand mathematical topics outside of their usual high school curriculum, as well as research and record a podcast about a famous female mathematician.

Podcasts



The Women in Tech Show: A Technical Podcast [LISTEN NOW](#)

A podcast that features the great work that women do in tech! The Women in Tech Show: A Technical Podcast is a show about the tech being developed by real women in tech, what they are working on and what they have worked on the past, and some tips on career development.



People Behind the Science [LISTEN NOW](#)

Explore the lives and experiences of the people behind the research and scientific discoveries of today.



The Random Sample [LISTEN NOW](#)

The Random Sample will give you a unique glimpse into the world of mathematics and statistics, and how these sciences play such a crucial role in so many areas where you wouldn't expect.



Ordinarily Extraordinary – Conversations with women in STEM

The podcast features different women working in different STEM fields and are at different phases in their careers. Through this podcast, women prove that they too can bring their unique perspectives and stories in STEM.

[LISTEN NOW](#)


Science Vs [LISTEN NOW](#)

Science Vs takes on fads, trends, and the opinionated mob to find out what's fact, what's not, and what's somewhere in between.



Flash Forward [LISTEN NOW](#)

Flash Forward is a critically acclaimed podcast about the future. In each episode, host Rose Eveleth takes on a possible (or not-so-possible) future scenario—everything from the existence of artificial wombs, to what would happen if space pirates dragged a second moon to Earth.



Videos and Documentaries

The Knowledge Channel

[ACCESS NOW](#)

The Knowledge Channel Foundation Inc. (KCFI) is a non-stock and non-profit organization established in 1999, and is the first and only curriculum-based educational TV channel in the Philippines. Following the Department of Education curriculum as a guide, KCFI brings K-12 and Alternative Learning subjects and concepts to life on TV, the internet and video on demand in the classrooms, learning centers, and homes.

SHE-ensya

[ACCESS NOW](#)

SHE-ensya is a science blog and vlog hosted by Dr. Chona Vince Cruz-Abeledo, a Filipina scientist and educator. The channel presents bite-sized content and relevant information on STEM in the Philippines in a way is understandable and fun!

Gross Science

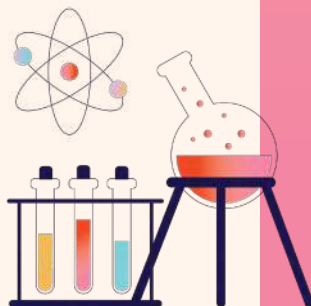
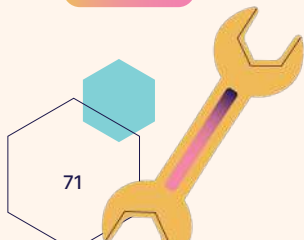
[ACCESS NOW](#)

Gross Science is a STEM YouTube channel created and hosted by Anna Rothschild. It covers everything “gross” about science—from what it’s like to eat bugs, and parasites in dinosaur poop. The channel is surely a little icky but definitely entertaining!

BrainCraft

[ACCESS NOW](#)

All your brain-based science and psychology questions are answered in these fun, colourful, and energetic videos hosted by STEM YouTuber Vanessa Hill of BrainCraft.





Videos and Documentaries

SciShow

[ACCESS NOW](#)

SciShow delves into the scientific subjects that defy our expectations and make us even more curious about STEM! Students can ask questions, contribute to a live debate, or simply sit back and enjoy how the hosts deconstruct the weird things that happen in our world.

Khan Academy

[ACCESS NOW](#)

Khan Academy creates software, videos, and other forms of content designed to help teachers (and parents) improve the delivery of STEM content and lessons to their students.

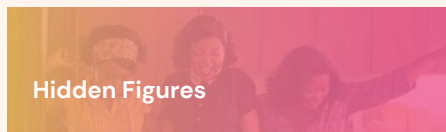


The Coding Train

[ACCESS NOW](#)

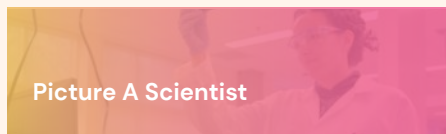
The Coding Train has video tutorials on a wide range of coding topics, from the basics of JavaScript to algorithmic art, machine learning, simulation, generative poetry, and more.

Hidden Figures



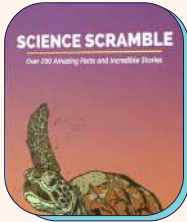
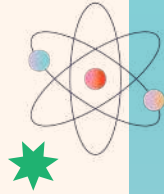
In the 1960s, three African American female mathematicians were working on the technology that would later see the first human expedition into outer space. The problem? These women never received the recognition that their white male colleagues did. This STEM movie is based on their story.

Picture A Scientist



This is a documentary about female scientists who talk about how they've had to deal with unfair treatment as they write a new chapter in STEM for women.

Reading Resources



Science Scramble:
Over 200 Amazing
Facts and
Incredible Stories
FlipScience

An illustrated book that contains fun facts, surprising stories, and myth-busters, all related to science. You can feed your curiosity and expand your mind by picking up this book of science stories and facts.



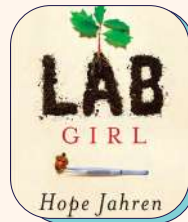
Istoryang
Siyentipiko: A
Showcase of
Filipino Scientific
Excellence
FlipScience

A compilation of Pinoy scientist features and profiles in an easily accessible format that anyone, anywhere can enjoy and draw inspiration from.



In the Land of
Invisible Women:
A Female Doctor's
Journey in the
Saudi Kingdom
Qanta A. Ahmed

This memoir by Qanta A. Ahmed, a British Muslim doctor, begins in the late 1990s when her application to renew her U.S. visa is denied and she accepts a job in Saudi Arabia. Though excited about the prospect of forming a deeper connection with her faith, she struggles with being a feminist, a doctor, and a Western woman living and working in a country that's deeply oppressive to women.



Lab Girl
Hope Jahren

Hope Jahren chronicles her lifelong passion for nature and science. Her memoir delves into her relationship with her parents and the role it played in cultivating her love of science as a child, as well as how her passion for plants and science helped her gain a better understanding of herself.

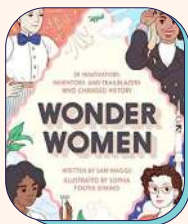


Under the Stars:
Astrophysics for
Bedtime
Lisa Harvey-Smith

Lisa Harvey-Smith is an Astronomer, Author and the Australian Government's Women in STEM Ambassador. Her children's book, *Under the Stars: Astrophysics for Bedtime* takes children and adults on an incredible journey through the night sky.



Reading Resources



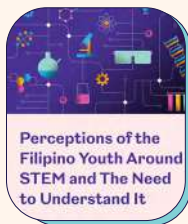
**Wonder Women:
25 Innovators,
Inventors, and
Trailblazers Who
Changed History**
Sam Maggs

Sam Maggs, a writer who frequently explores what it means to be a woman in geek culture, uses illustrated biographies to examine the lives of scientists, engineers, inventors, and more. With a little humor, references to pop culture, and a conversational tone, Maggs shows the amazing things that women have done in history.



**Women in Science:
50 Fearless
Pioneers Who
Changed the World**
*Rachel
Ignotofsky*

Women in Science highlights the contributions of fifty notable women to the fields of science, technology, engineering, and mathematics (STEM) from antiquity to the modern world. Full of striking, singular art, this collection also contains infographics about relevant topics such as lab equipment, the rates of women currently working in STEM fields, and an illustrated scientific glossary.



**Perceptions of
the Filipino Youth
Towards STEM
and The Need to
Understand It**

[READ MORE](#)

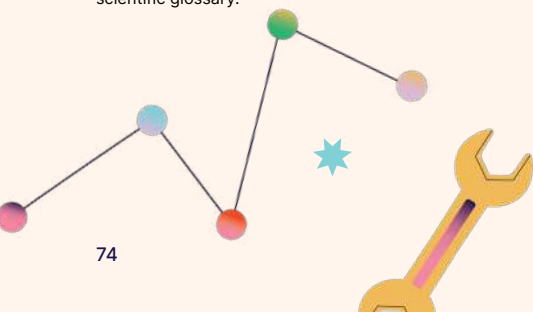
STEM+PH, together with the Philippine Business Coalition for Women Empowerment (PBCWE) and YouthInsight, the full service research arm of Student Edge, launched the Youth in STEM report to understand and determine the interests, perceptions, and even barriers in STEM education and careers between male and female students. Through this, we hope to help them better understand their own future, as well as the future of humanity.



**Asian Scientist
Magazine**

[WEBSITE](#)

Asian Scientist Magazine is an award-winning science and technology magazine that highlights R&D news stories from Asia to a global audience. The magazine is published by Singapore-headquartered Wildtype Media Group, Asia's leading STEM and healthcare media company. Wildtype Media Group also publishes books and Supercomputing Asia, a print magazine focusing on the high performance computing sector.



Reading Resources

From the Pinays Can STEM blog!



The boss, not bossy: 9 STEM Women taking the lead

This time around, we're introducing even more women advocating for a more gender-inclusive future for all. Through their programs and initiatives, these women prove that girls aren't bossy; we're just the boss!



Stand With Us, Not For Us: How men can be better allies

It's not just girls needing each other's support! Here are five ways men can stand in solidarity with women—because gender equality extends to STEM and beyond.



Like Mother, Like Daughter: Meet the STEMoms and daughters!

We're doubling up our STEM stories with these inspiring mother-daughter pairs in STEM. They share how they started out in the field, how to raise a daughter in STEM, and, of course, their own Mother's Day dedication cards.



STEM on Site: What it takes to be a woman on the field

STEM doesn't just happen in the confines of your room or inside a lab! We peek through the Field Notes of these STEM women on the field—and find out what happens right where the action is!



Our Sheroes: The First Pinays of STEM

Throughout history, women have long been cast in the shadows of their male counterparts—and the same can be said for our Filipina STEM pioneers. Like any worthy superhero, we're here to get to know their awe-inspiring origin stories—and what we can learn from our STEM founding mothers.

Home Experiments



Black Magic

Find out the secret colors found in a black marker!

Naked egg

Put de-shelled eggs in different fluids and watch them swell and shrink.

Paper airplanes

Build, test, and improve paper airplanes.

How do things fall?

Find out what determines how quickly objects fall.

Household energy audit

Review the electrical appliances used at home.

Design a better vortex cannon

Build vortex cannons out of household materials.

Cloud in a Jar

Make your own cloud using objects at home!

Sticky Note Number Match

Understand the meaning of numeric values through this movement-oriented activity.

Collecting Light

Learn how light and energy are spread throughout space!

Cotton Ball Launcher

Build a rubber band-powered cotton ball launcher.

Make your own eggshell chalk

Yes, you can make chalk out of eggshells!

Secret Message with Lemon Juice

Lemon juice can be used as ink for secret messages!

Electricity Circuits

Experiment with batteries, voltages and light bulbs.

Build a Pizza Box Solar Oven

Gather the sun's rays and cook a tasty treat.

Egg Drop Project

Protect a raw egg from a high fall by using various materials.

Homemade Battery

A simple experiment to help you understand how batteries work!

Juice Box Geometry

Measure the dimensions of different juice boxes.

Measuring Tree Height

An easier way of measuring a tree!

Perfect Paper Cube

A great way to develop your ability to accurately read a ruler!

Bubble Geometry

Learn more about soap film properties and surface tension.

Seize the advocacy

Printables and online posters and templates

Join the fun online! Use these posters and become advocates of STEM yourselves!

DOWNLOAD



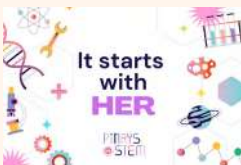
Cover photo



Profile picture frame



Pinays Can STEM poster



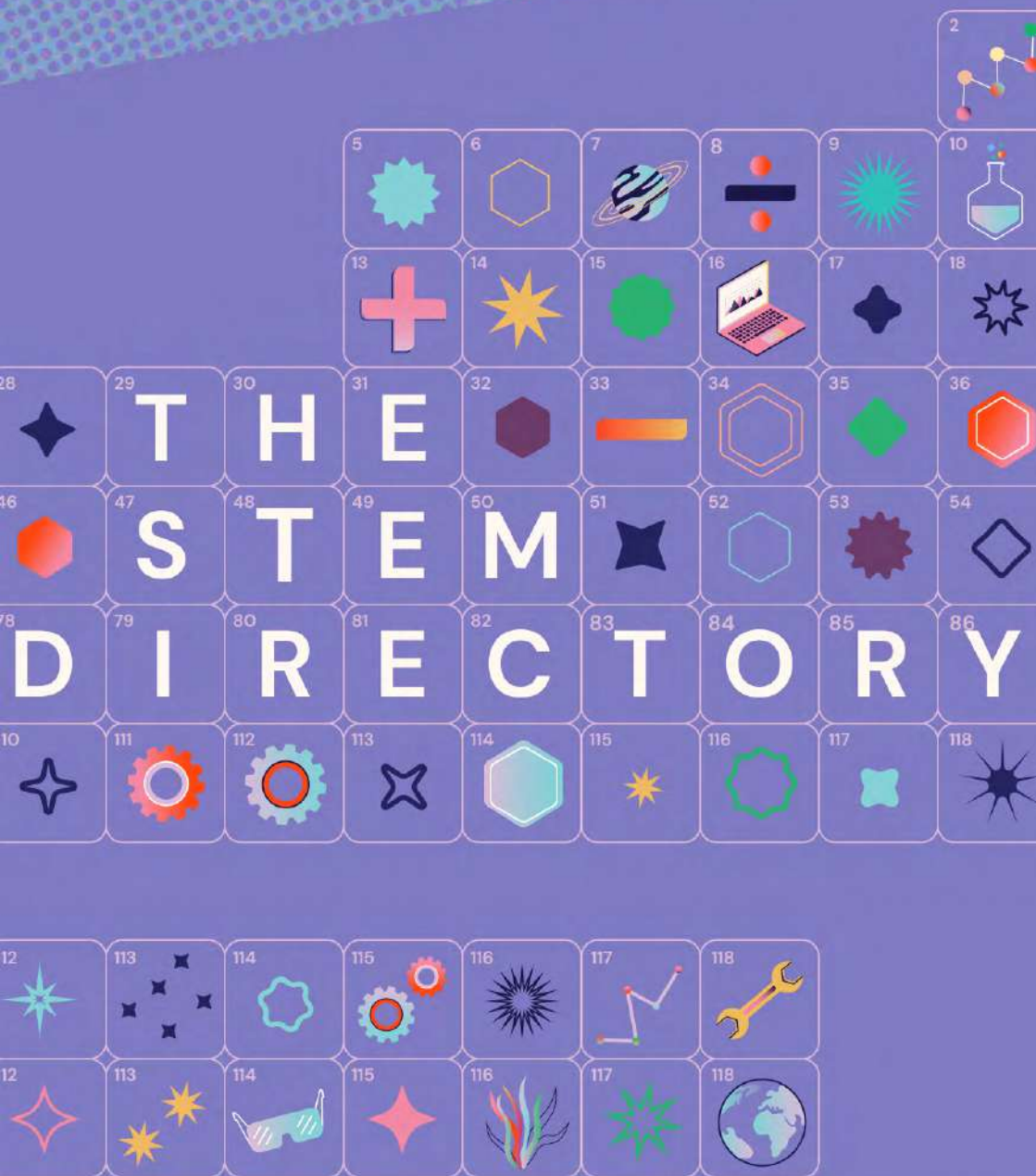
Printable poster

Desktop, phone, tablet wallpapers

Bring your inspiration wherever you go. Download these wallpapers for your phones, tablets, and computers!

DOWNLOAD





Women Empowerment



Girl Up Philippines

Girl Up Philippines is a United Nations Foundation's campaign that supports the empowerment of girls everywhere.



Philippine Commission on Women (PCW)

The Philippine Commission on Women is a government agency that works for the promotion of gender equality, women's empowerment, and women's rights.



Philippine Business Coalition for Women Empowerment (PBCWE)

The Philippine Business Coalition for Women Empowerment was launched in March 2017, through the Investing in Women initiative, by the Australian Government in partnership with the Philippine Women's Economic Network (PhilWEN).



Gender advocacy & STEM



STEM+PH

STEM+PH is the flagship program of Unilab Foundation, which aims to expand the pool of STEM Learners and Innovators in the country.



WiTech PH

WiTech PH (Women in Technology Philippines) is a nonprofit organization that aims to educate, inspire, and empower the Filipino youth to break gender barriers and use technology to make a difference in society. The organization now operates in more than 20 chapters in the Philippines and across the world.



Siyénsiyá PH

Siyénsiyá PH is a youth-led organization that aims to empower women and bridge societal gaps through STEM.



Girls Will Code PH

Girls Will Code PH is a community that encourages girls to participate in activities related to coding, programming, robotics, and engineering. Through this platform, girls can share ideas about their personal projects, promote interest in programming, and robotics in a fun learning environment.



FTW Foundation

For The Women Foundation (FTW) provides free data science and AI training for women.



International Labour Organization

Under ILO is the [Women in STEM Workforce Readiness & Development Programme](#), funded by the JPMorgan Chase Foundation. The programme seeks to provide women with critical soft and technical STEM-related skills, employability and leadership training coupled with targeted mentorship to help women gain quality employment and advancement opportunities in STEM-related jobs.

STEM Organizations



SciCreate

SciCreate is a youth initiative that aims to promote science communication and bridge the gap between researchers and the youth through the use of art, writing, and the humanities.



Association of Filipino Scientists in America

The Association of Filipino Scientists in America (AFSA) is a community of Filipino scientists that seek to utilize their expertise to support the Filipino community.



Developh

Developh is a community for mission-driven technologists in



iGEM Foundation

The iGEM Foundation is an independent, non-profit organization dedicated to the advancement of synthetic biology, education and competition, and the development of an open community and collaboration.



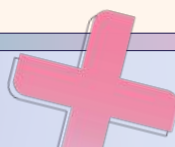
Pinoy Scientists

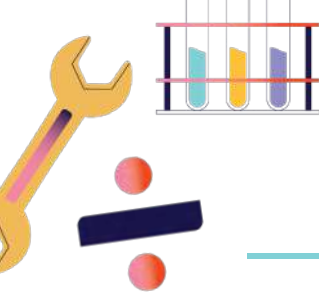
Pinoy Scientists is an online platform that showcases the different stories and testimonials of Filipinos in STEM all over the world!



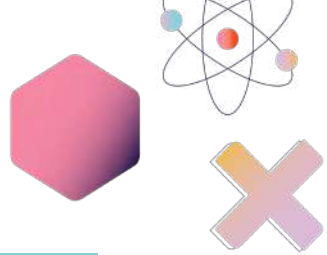
Center for Integrated STEM Education, Inc. (CISTEM)

Established in 2019 by Unilab Foundation, in partnership with the University of the Philippines Diliman College of Education, CISTEM aims to establish Integrated STEM Education and collaborates with like-minded organizations to revolutionize STEM Education in Asia.





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